

BFS Traversal

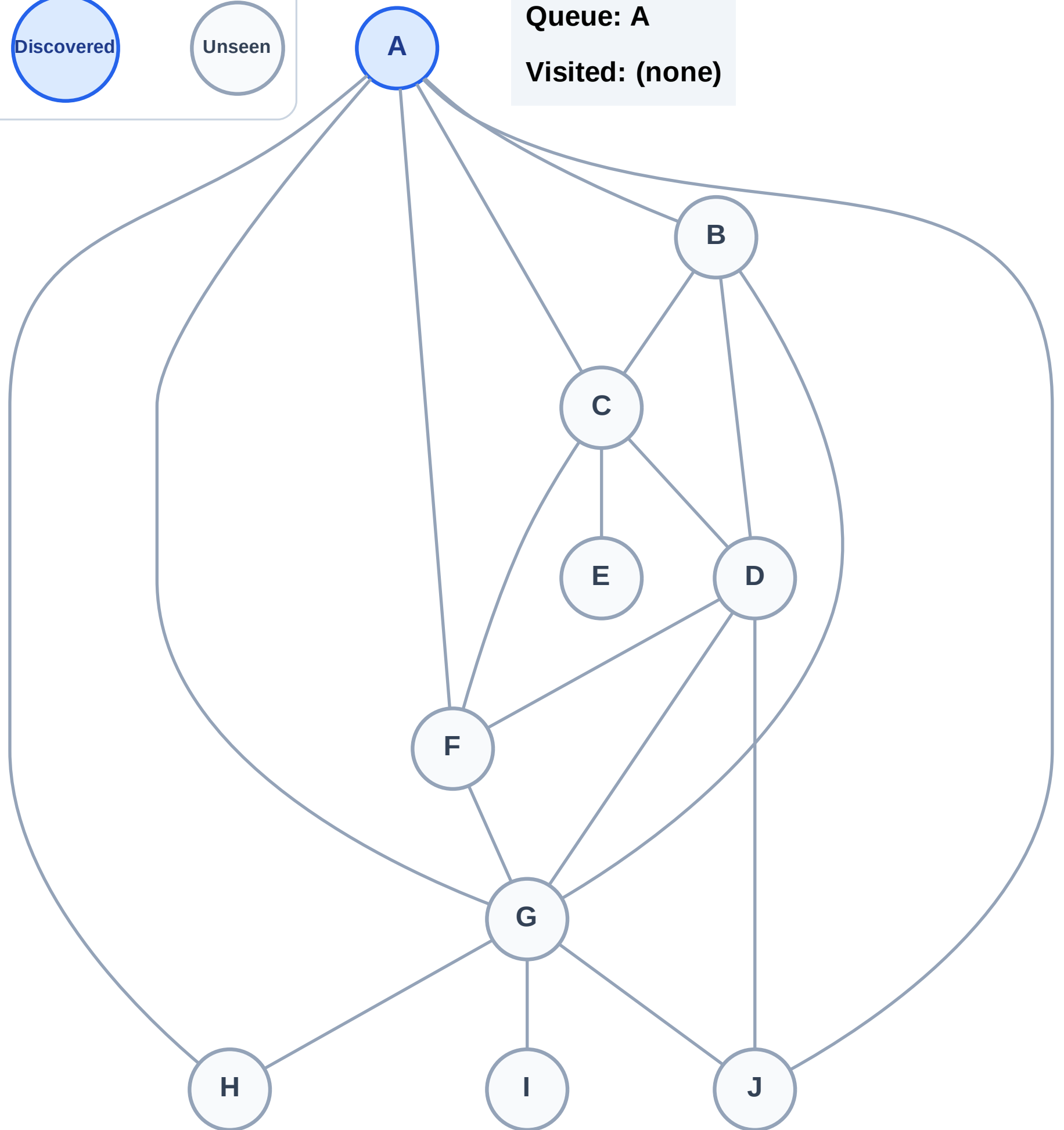
Step 1: Start at A

Legend



Queue: A

Visited: (none)



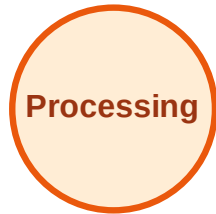
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



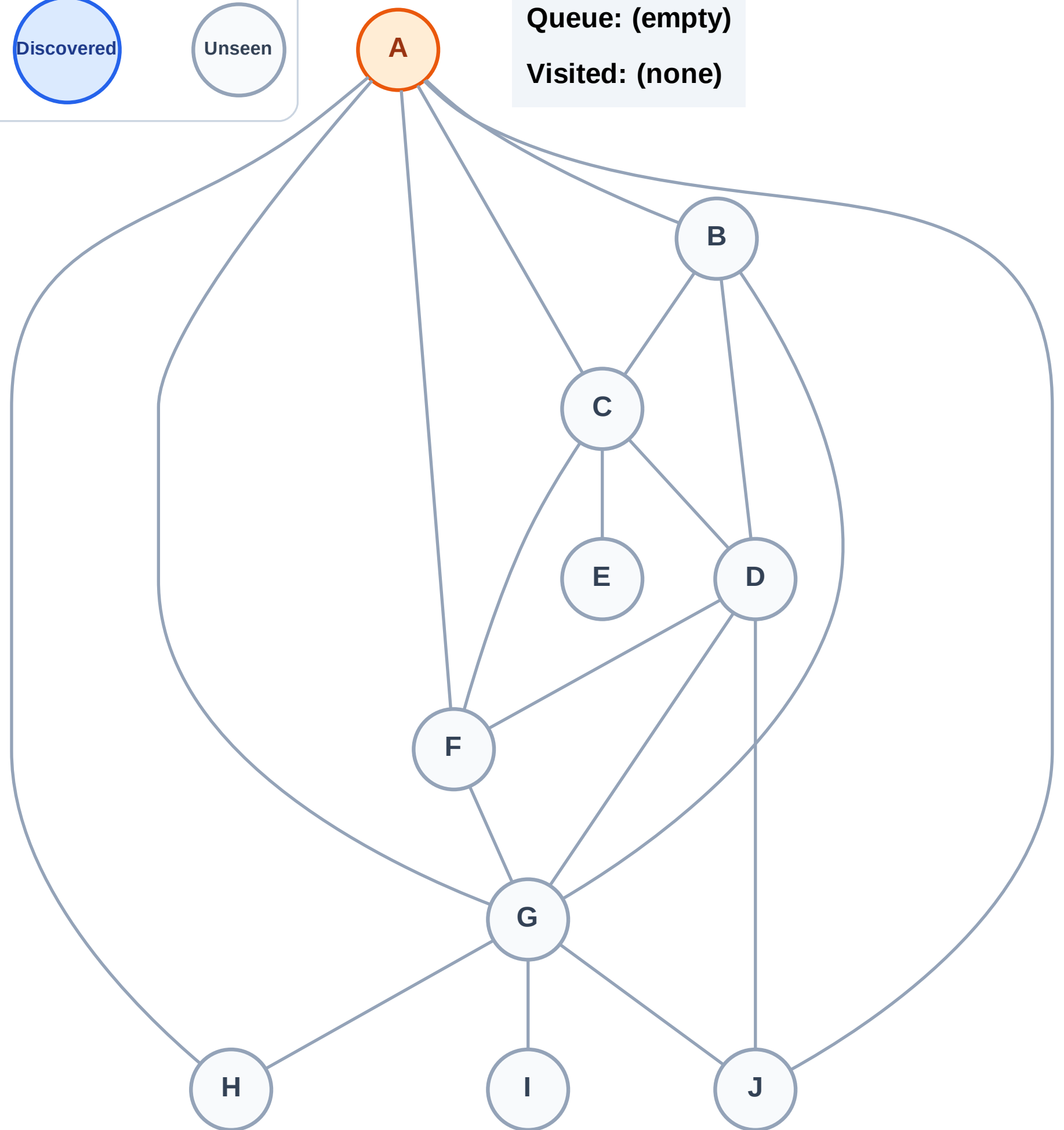
Discovered



Unseen

Queue: (empty)

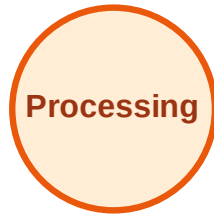
Visited: (none)



BFS Traversal

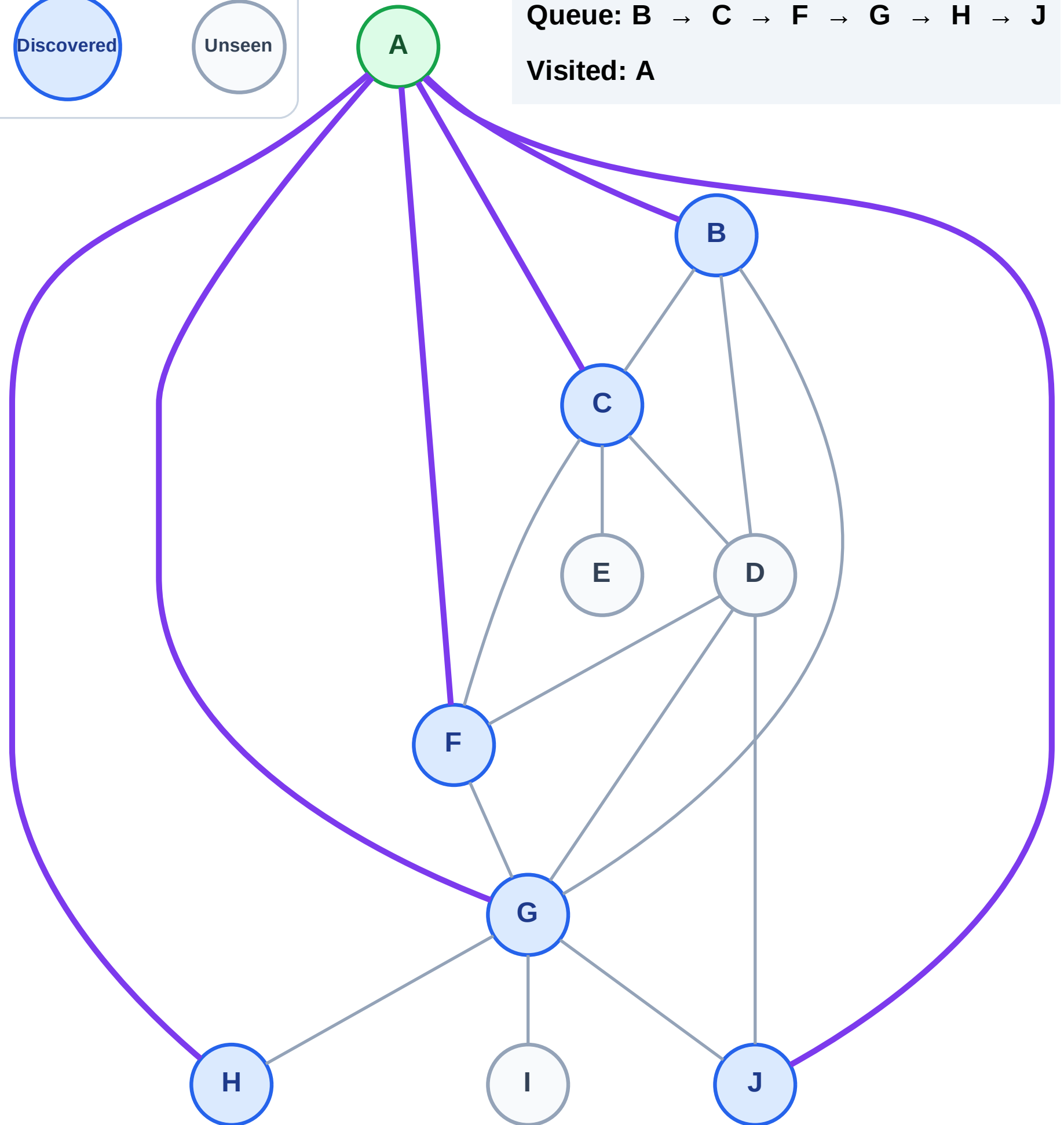
Step 3: Discover B, C, F, G, H, J from A

Legend



Queue: B → C → F → G → H → J

Visited: A



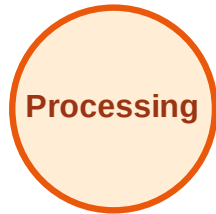
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



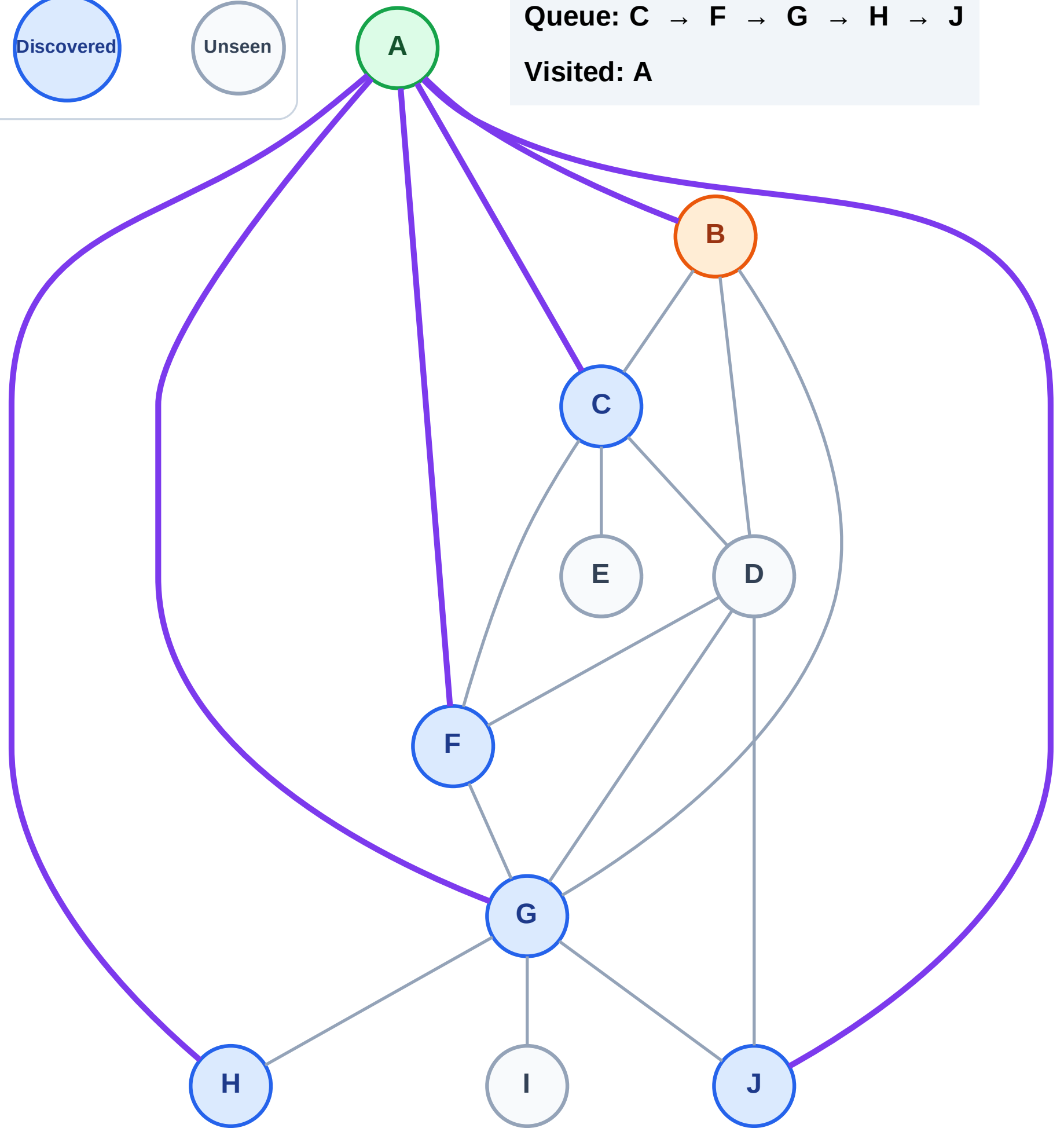
Discovered



Unseen

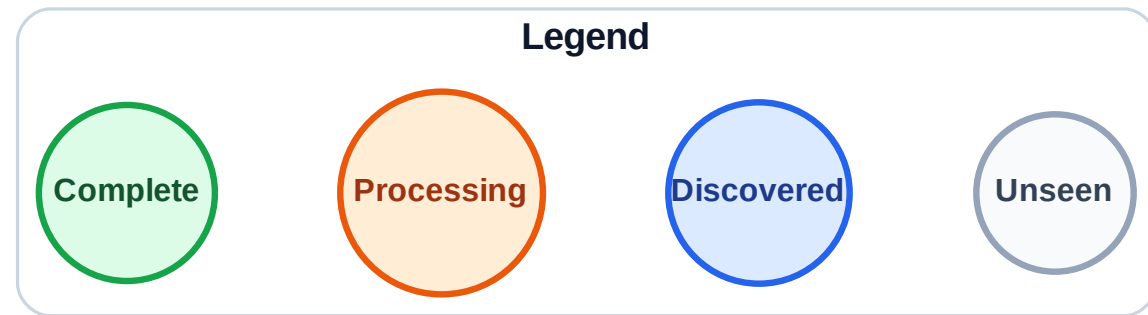
Queue: C → F → G → H → J

Visited: A



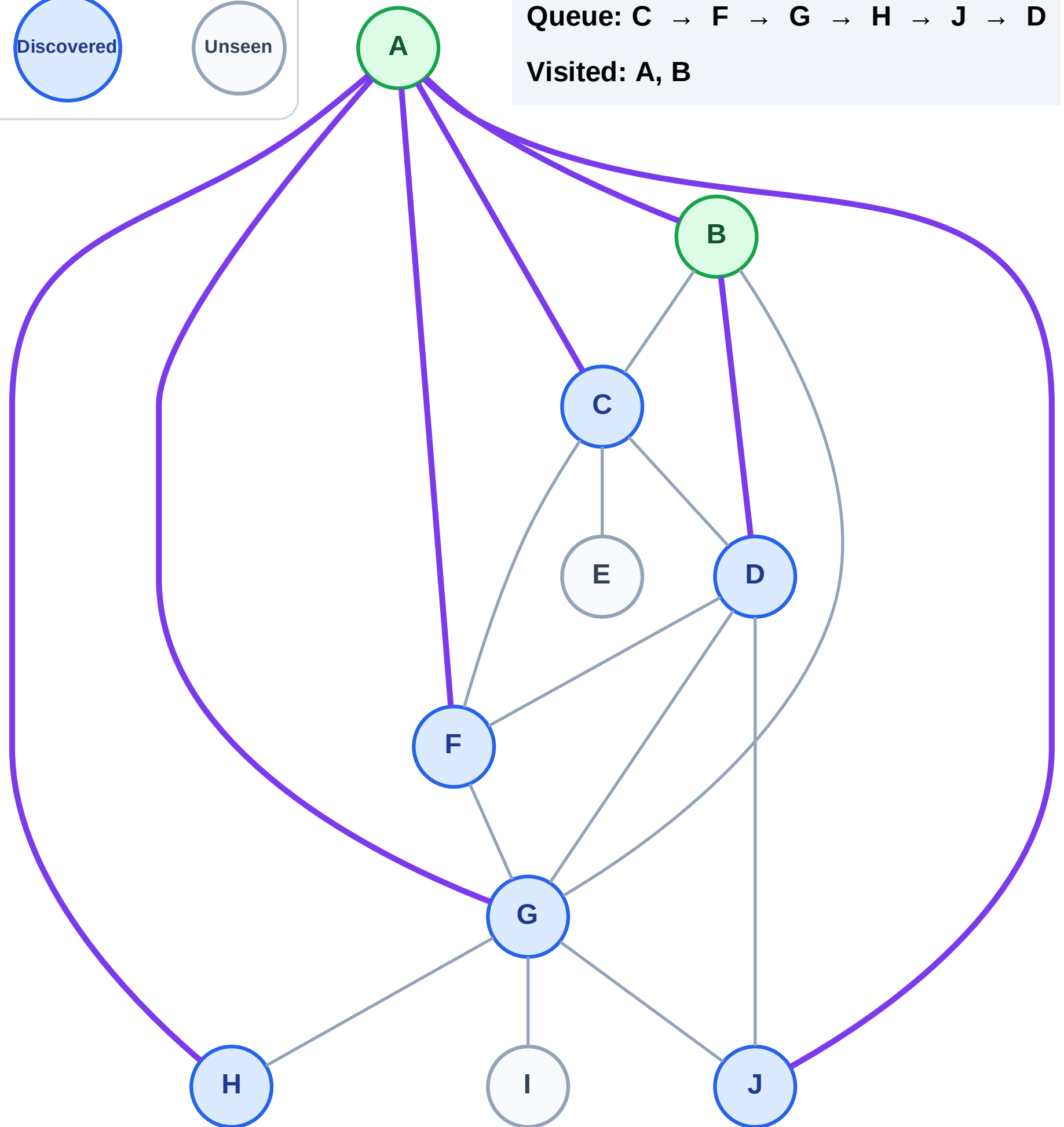
BFS Traversal

Step 5: Discover D from B



Queue: C → F → G → H → J → D

Visited: A, B



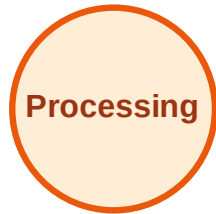
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



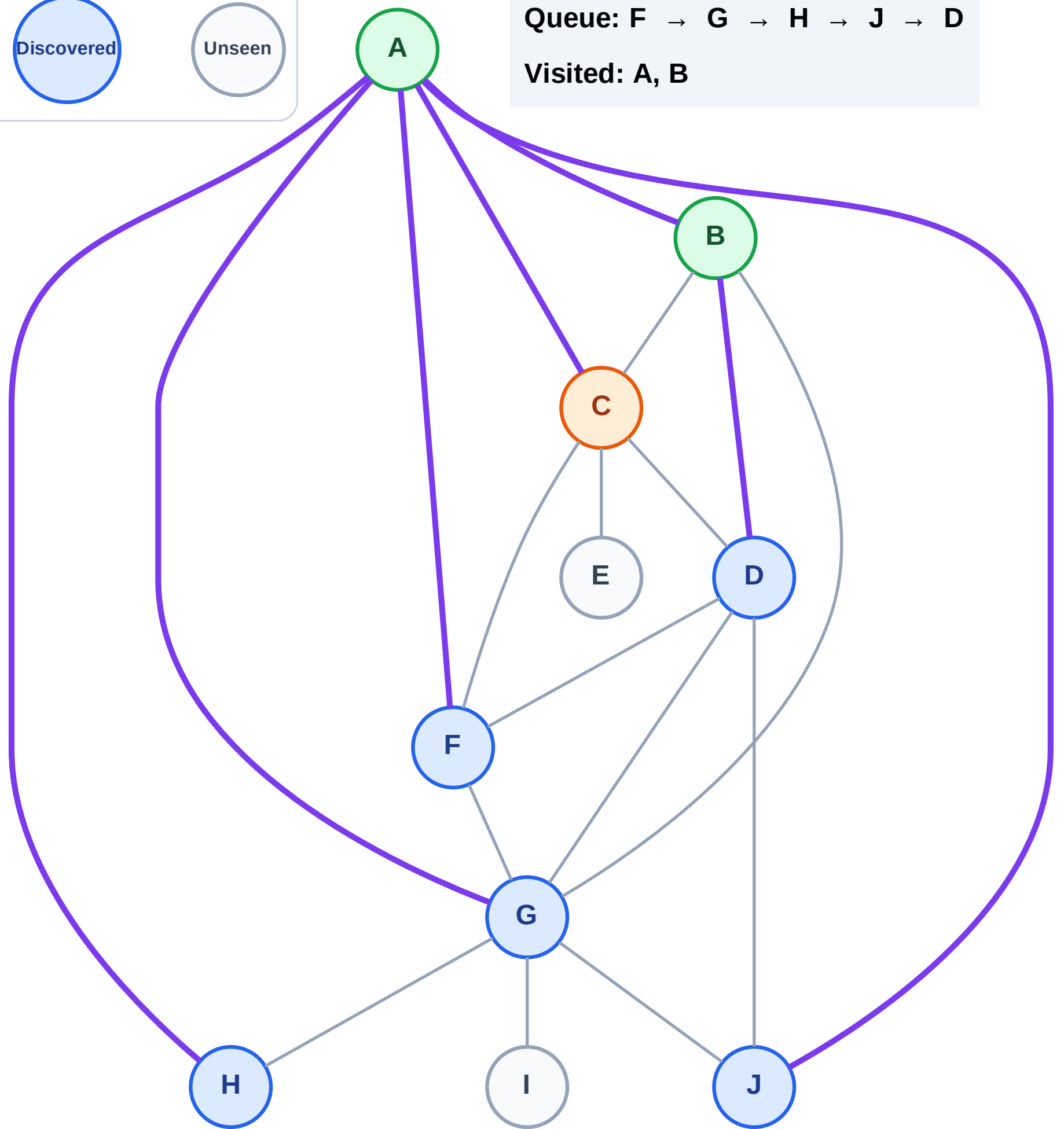
Discovered



Unseen

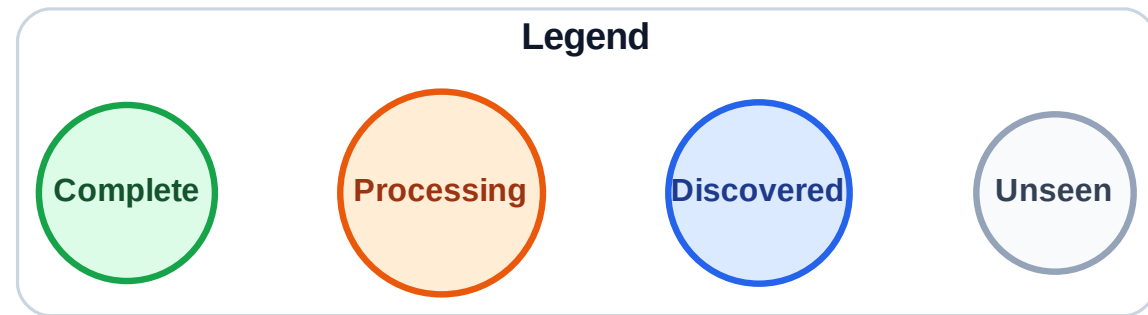
Queue: F → G → H → J → D

Visited: A, B



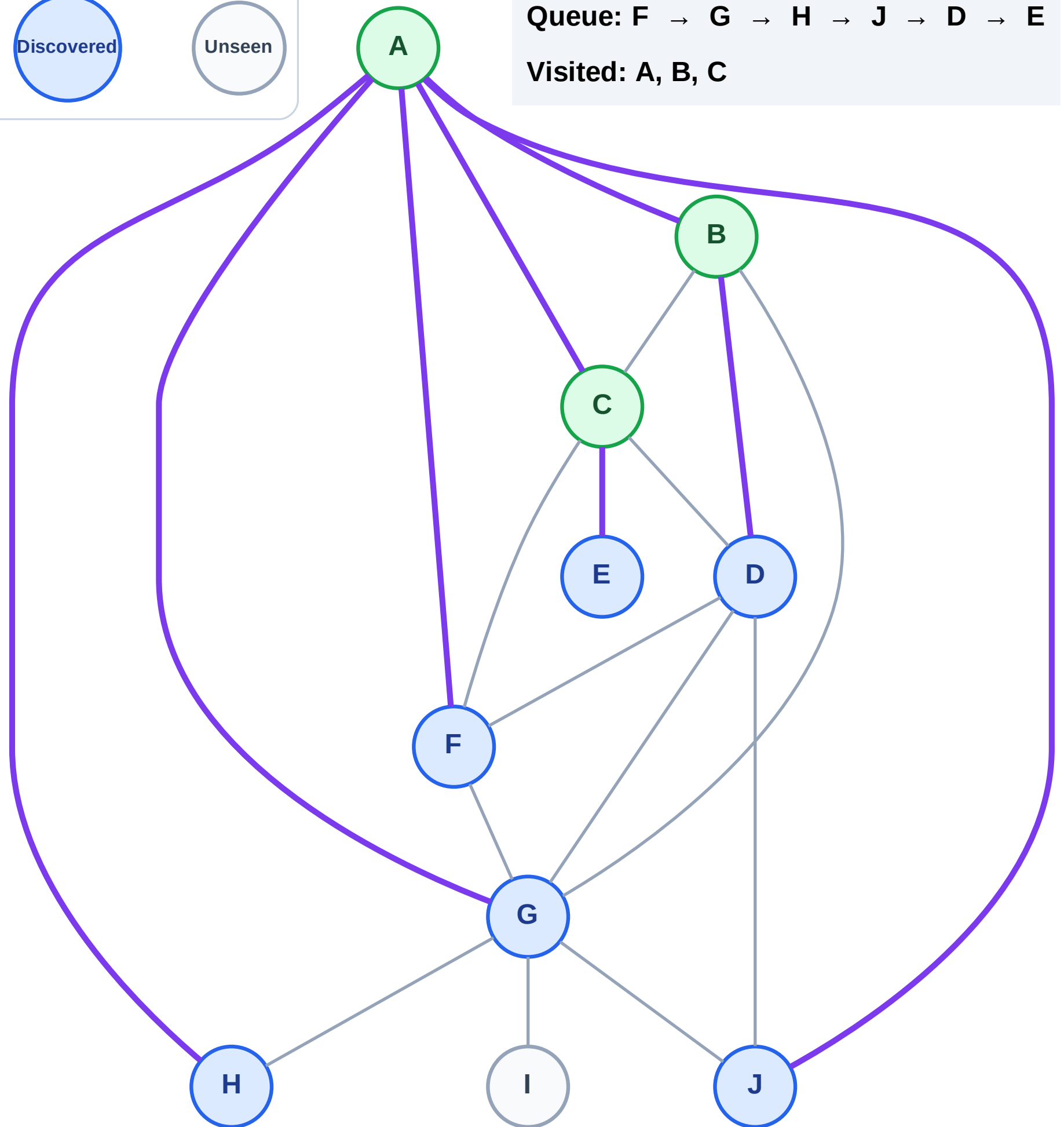
BFS Traversal

Step 7: Discover E from C



Queue: F → G → H → J → D → E

Visited: A, B, C



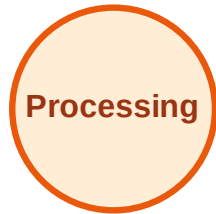
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



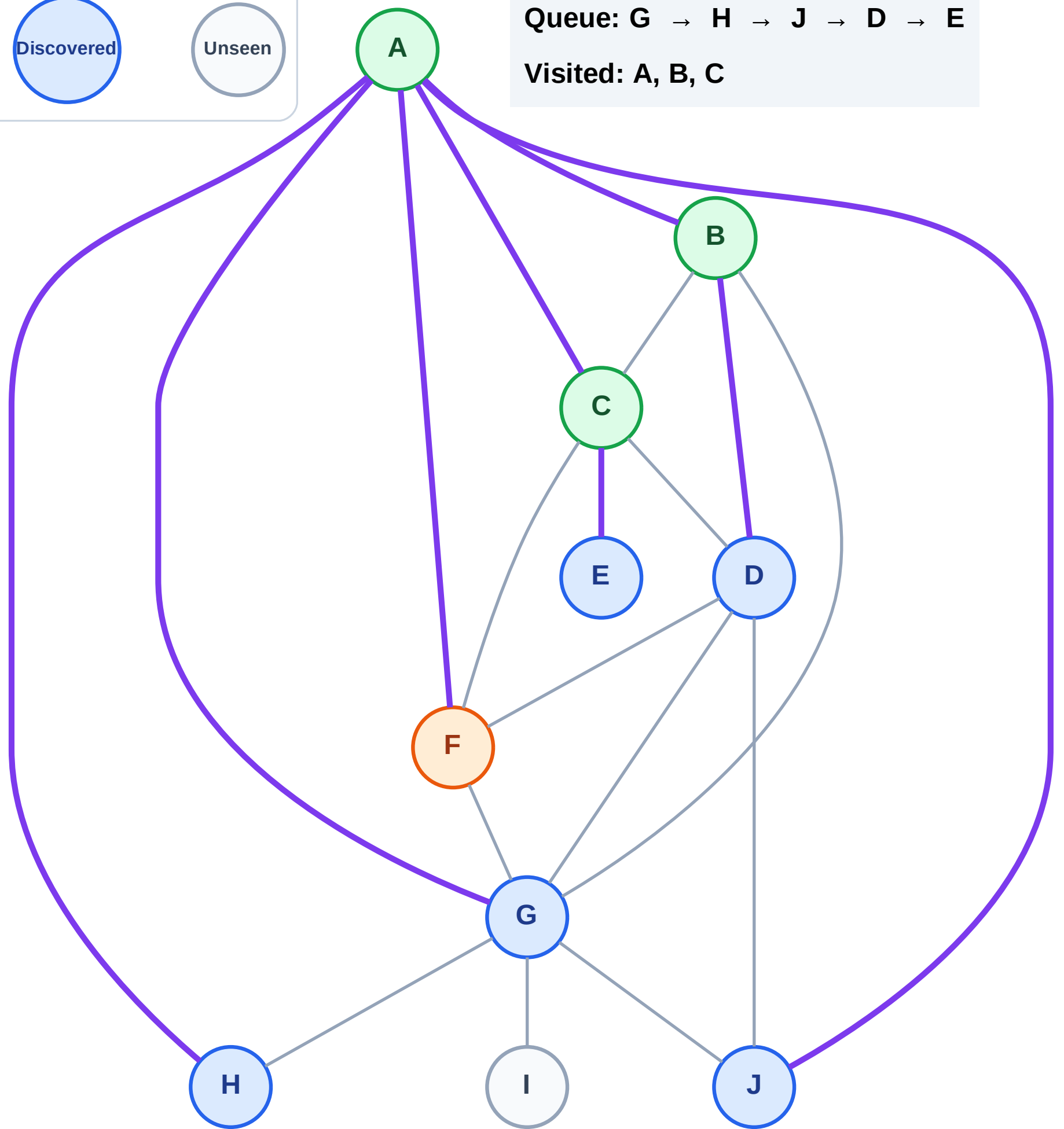
Discovered



Unseen

Queue: G → H → J → D → E

Visited: A, B, C



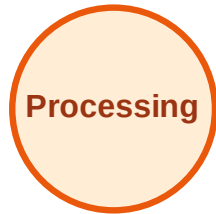
BFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



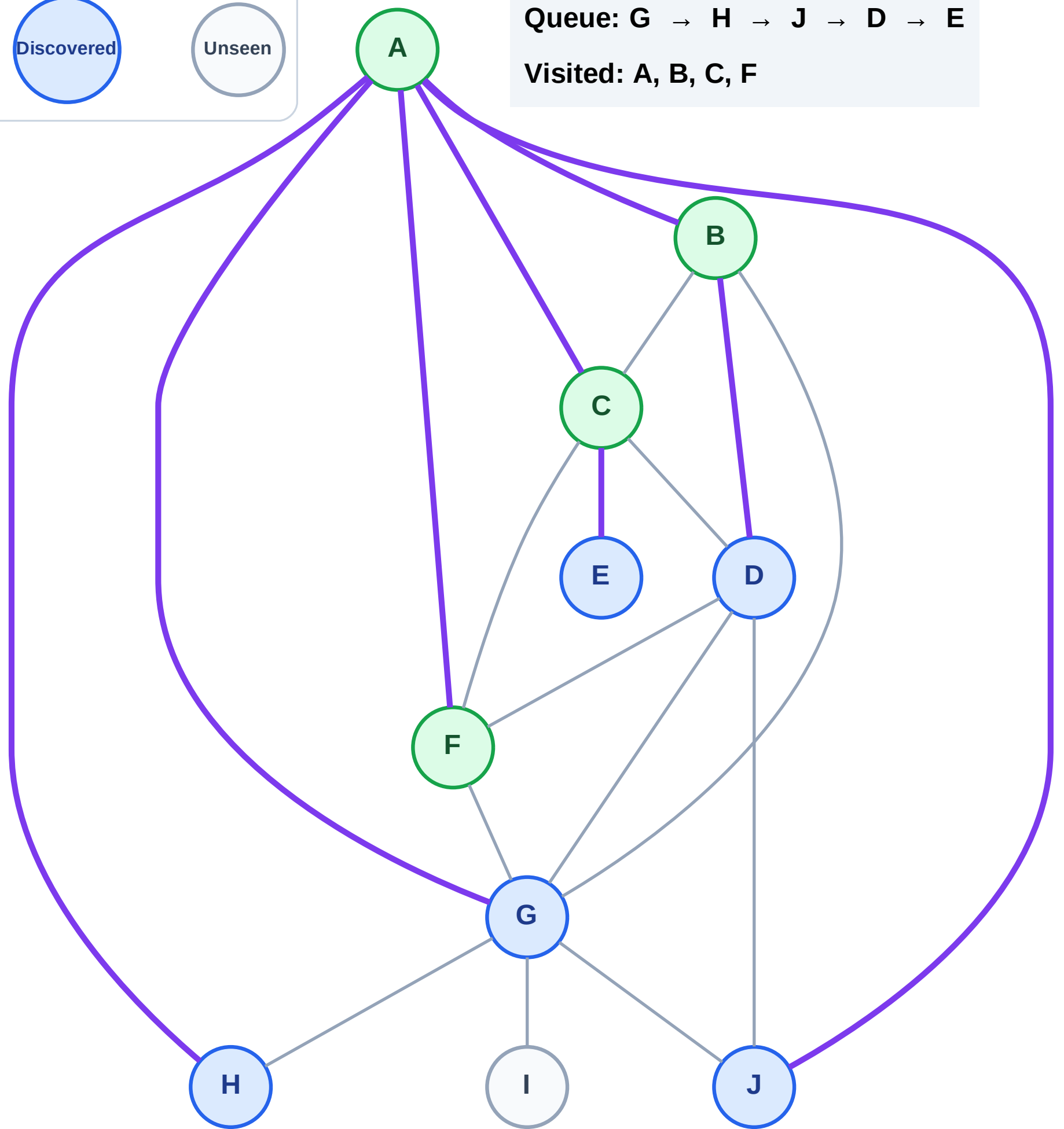
Discovered



Unseen

Queue: G → H → J → D → E

Visited: A, B, C, F



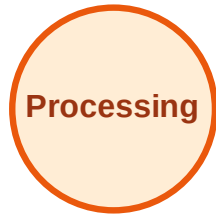
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



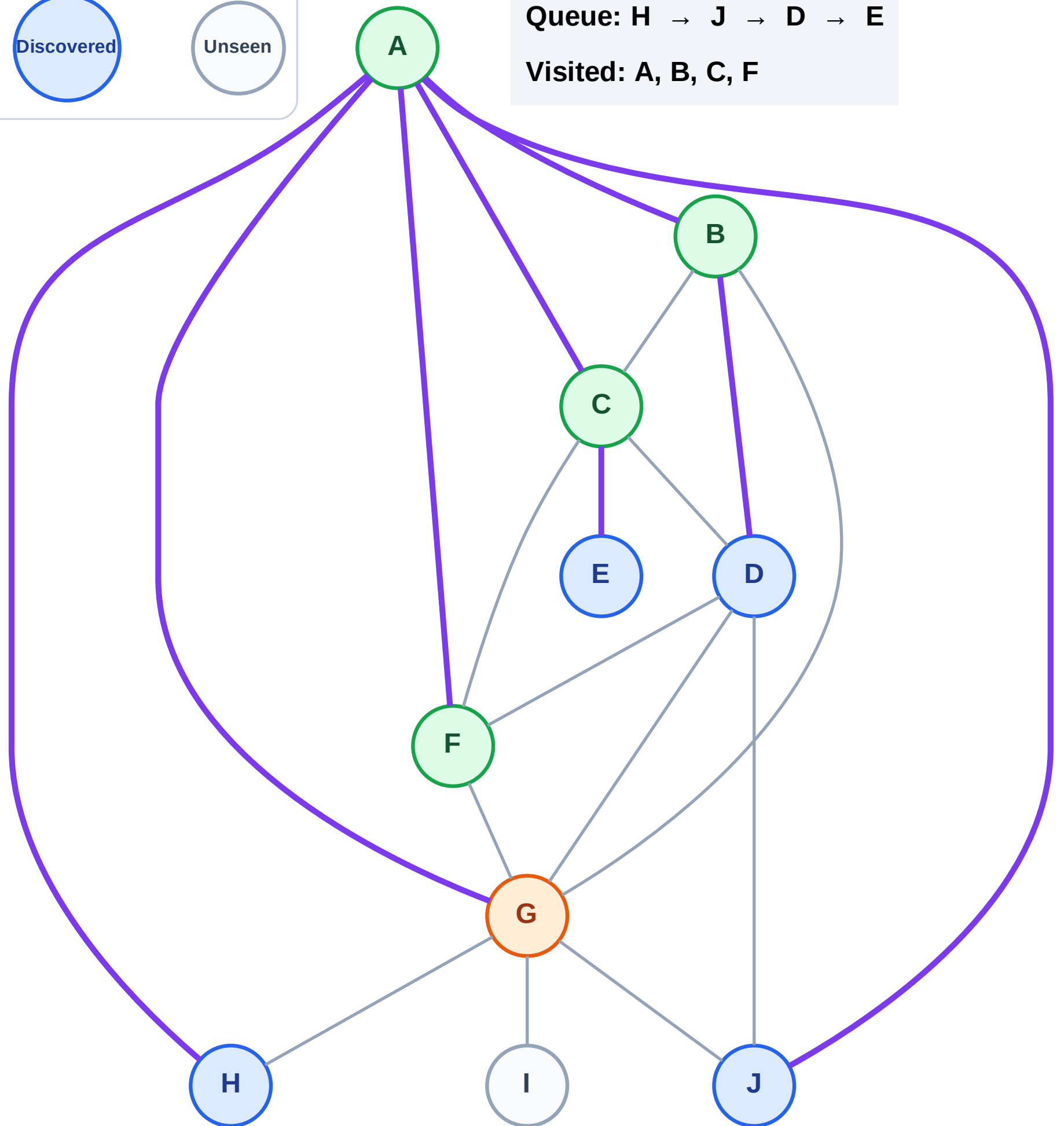
Discovered



Unseen

Queue: H → J → D → E

Visited: A, B, C, F



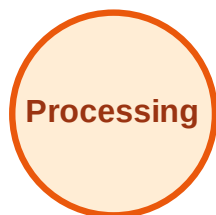
BFS Traversal

Step 11: Discover I from G

Legend



Complete



Processing



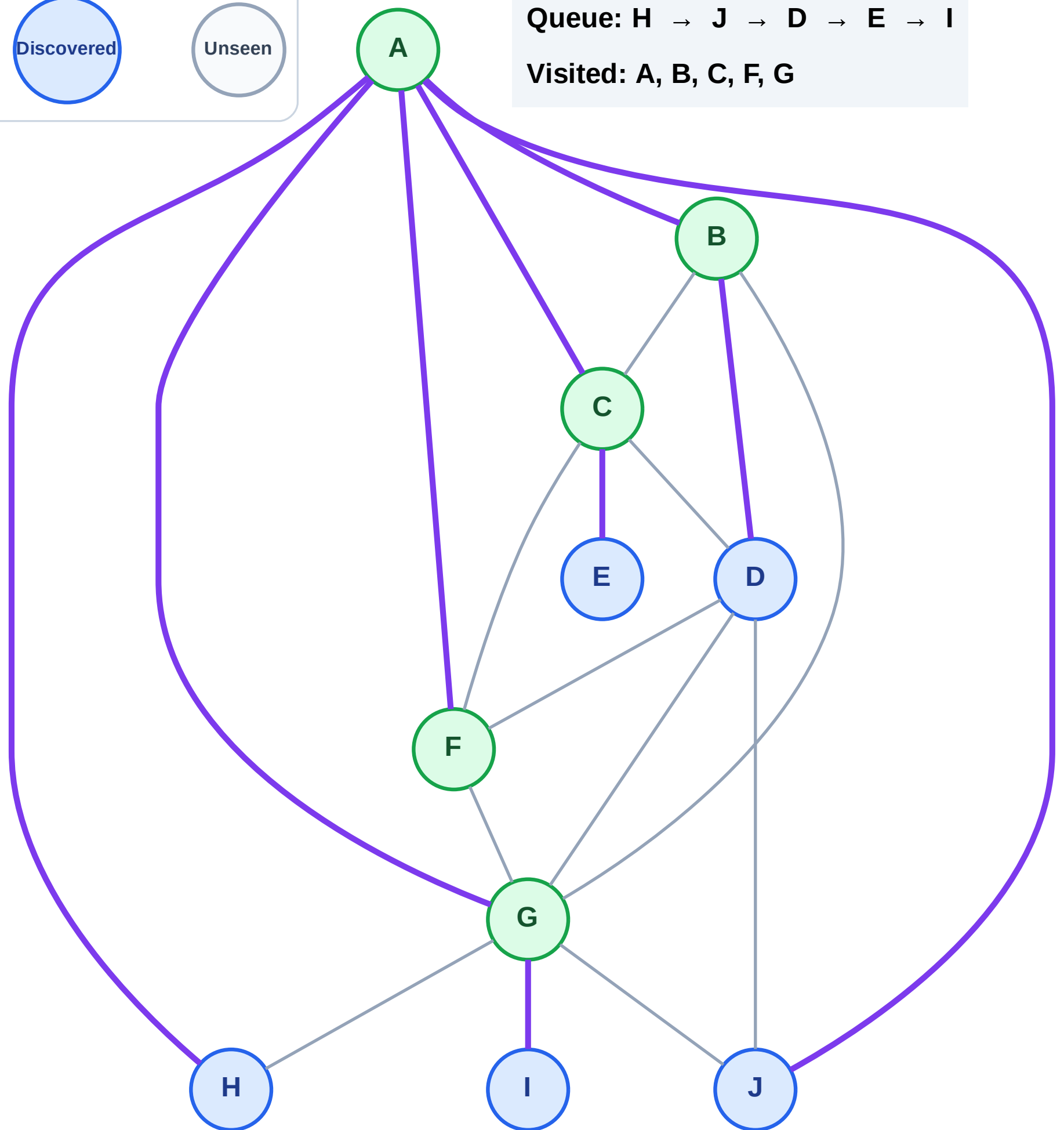
Discovered



Unseen

Queue: H → J → D → E → I

Visited: A, B, C, F, G



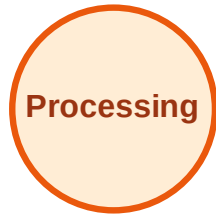
BFS Traversal

Step 12: Process node H

Legend



Complete



Processing



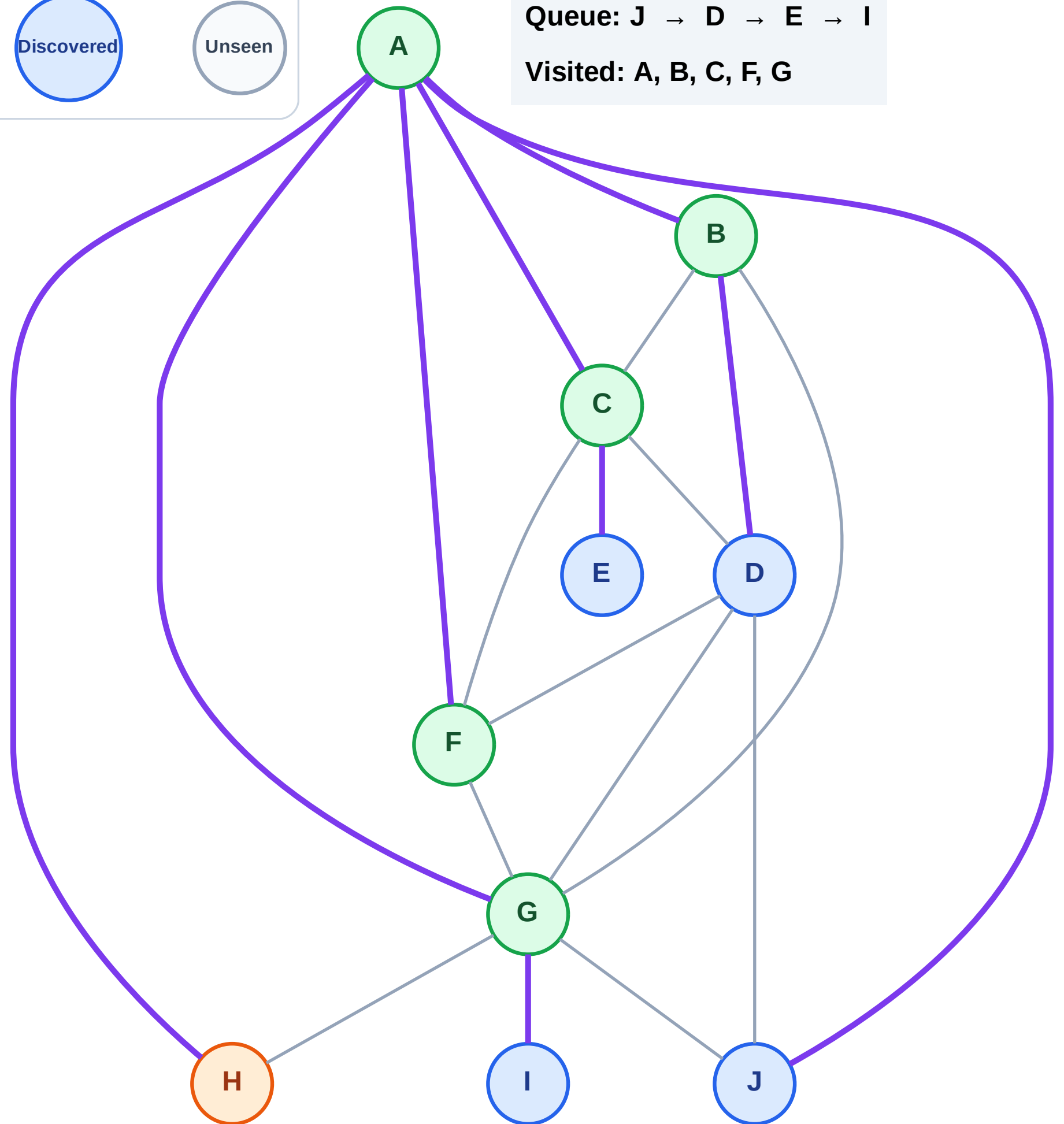
Discovered



Unseen

Queue: J → D → E → I

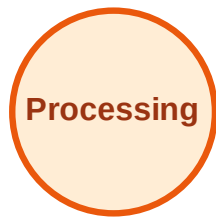
Visited: A, B, C, F, G



BFS Traversal

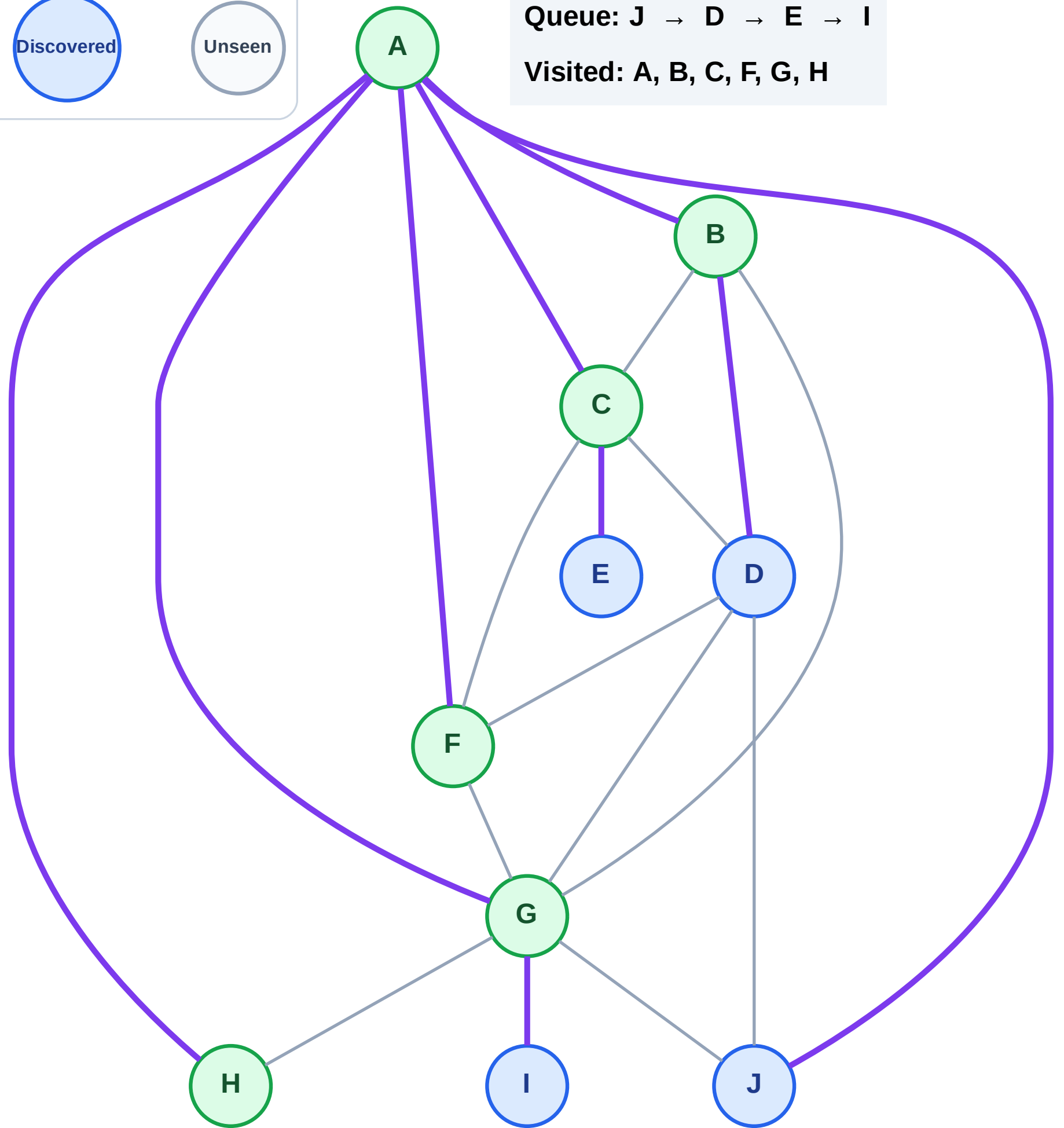
Step 13: No new neighbors from H

Legend



Queue: J → D → E → I

Visited: A, B, C, F, G, H



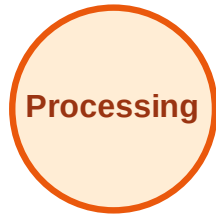
BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



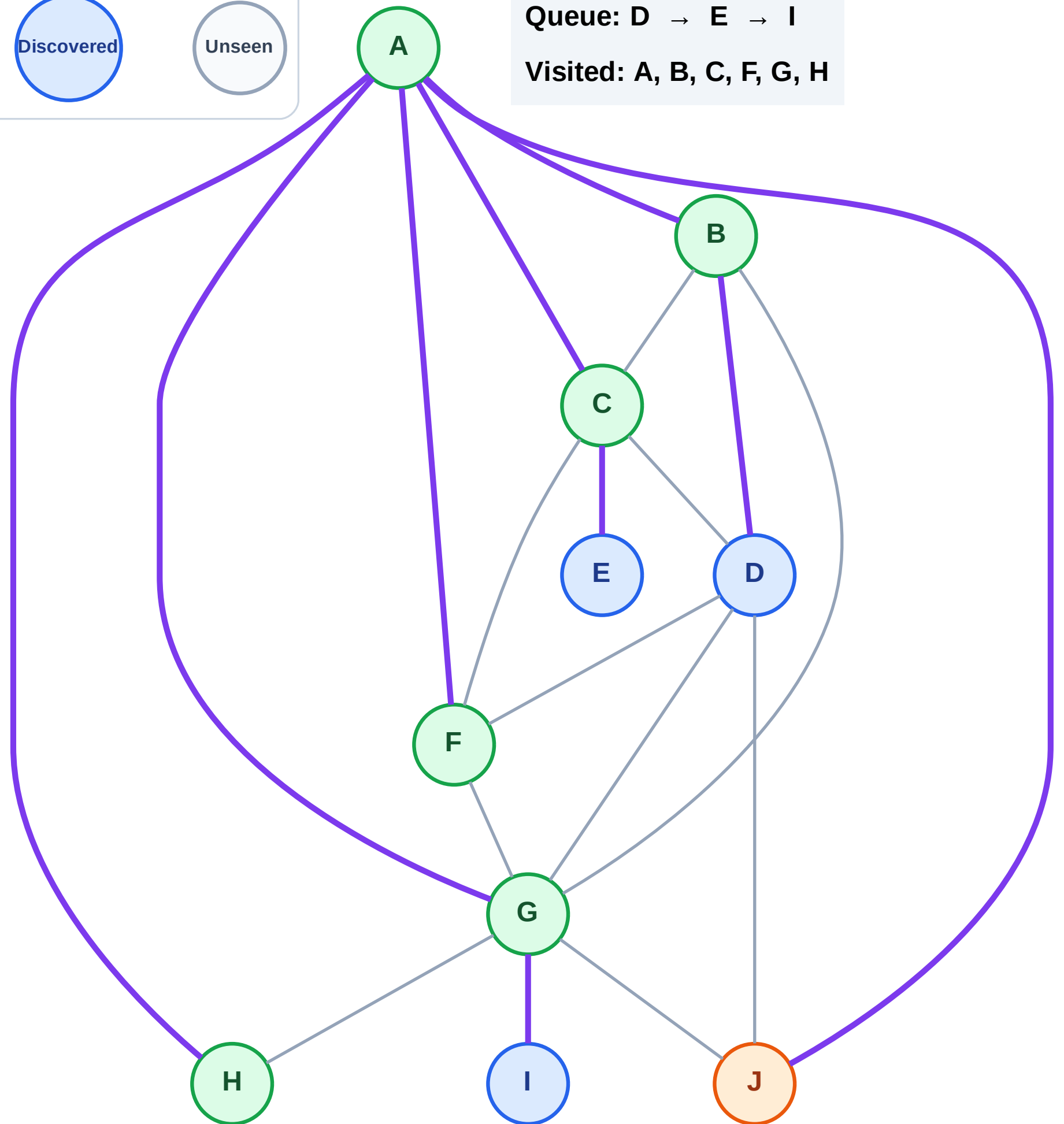
Discovered



Unseen

Queue: D → E → I

Visited: A, B, C, F, G, H



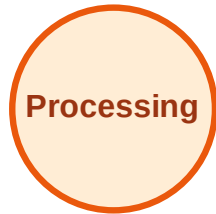
BFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing



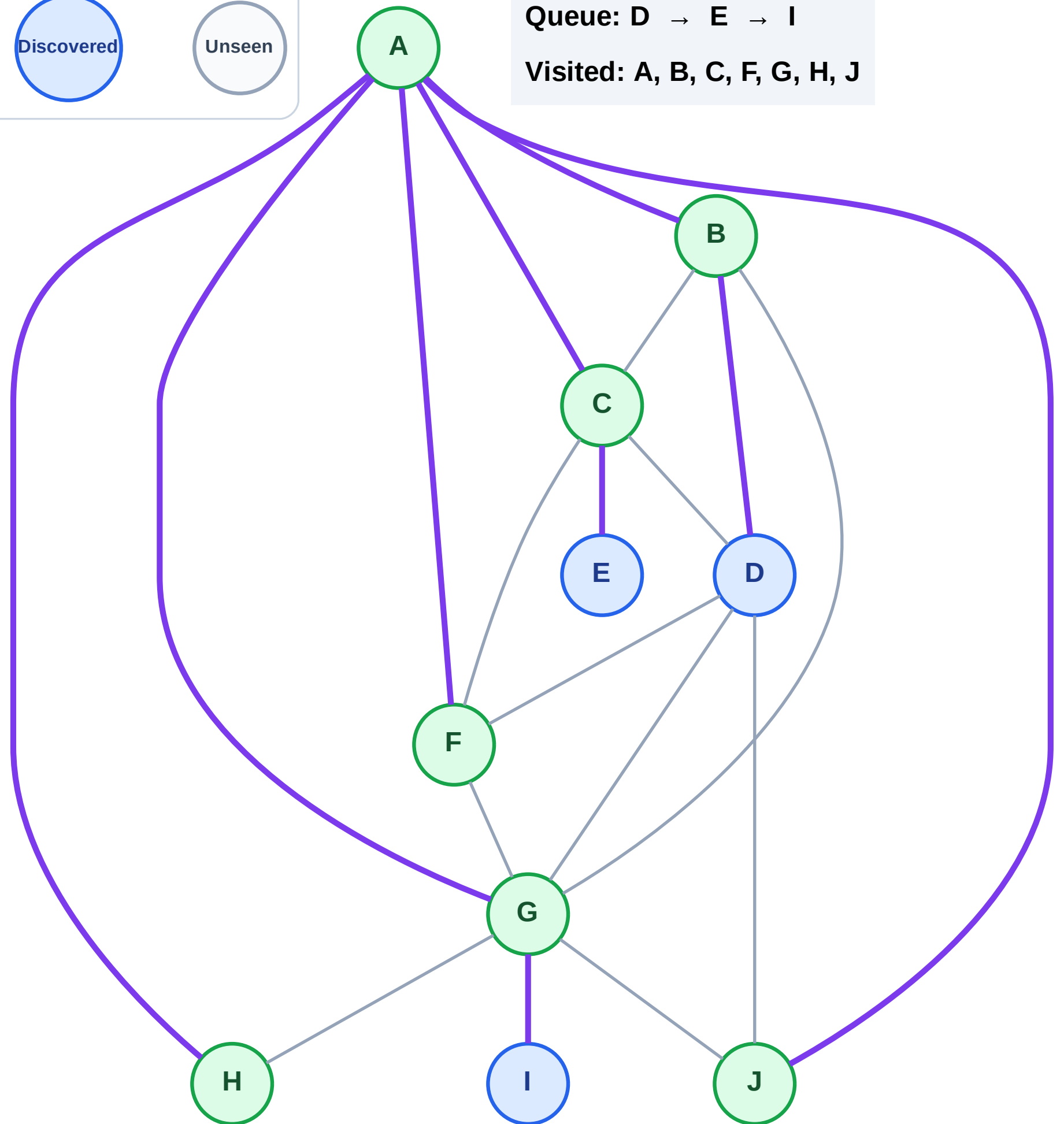
Discovered



Unseen

Queue: D → E → I

Visited: A, B, C, F, G, H, J



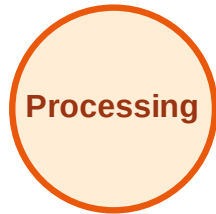
BFS Traversal

Step 16: Process node D

Legend



Complete



Processing



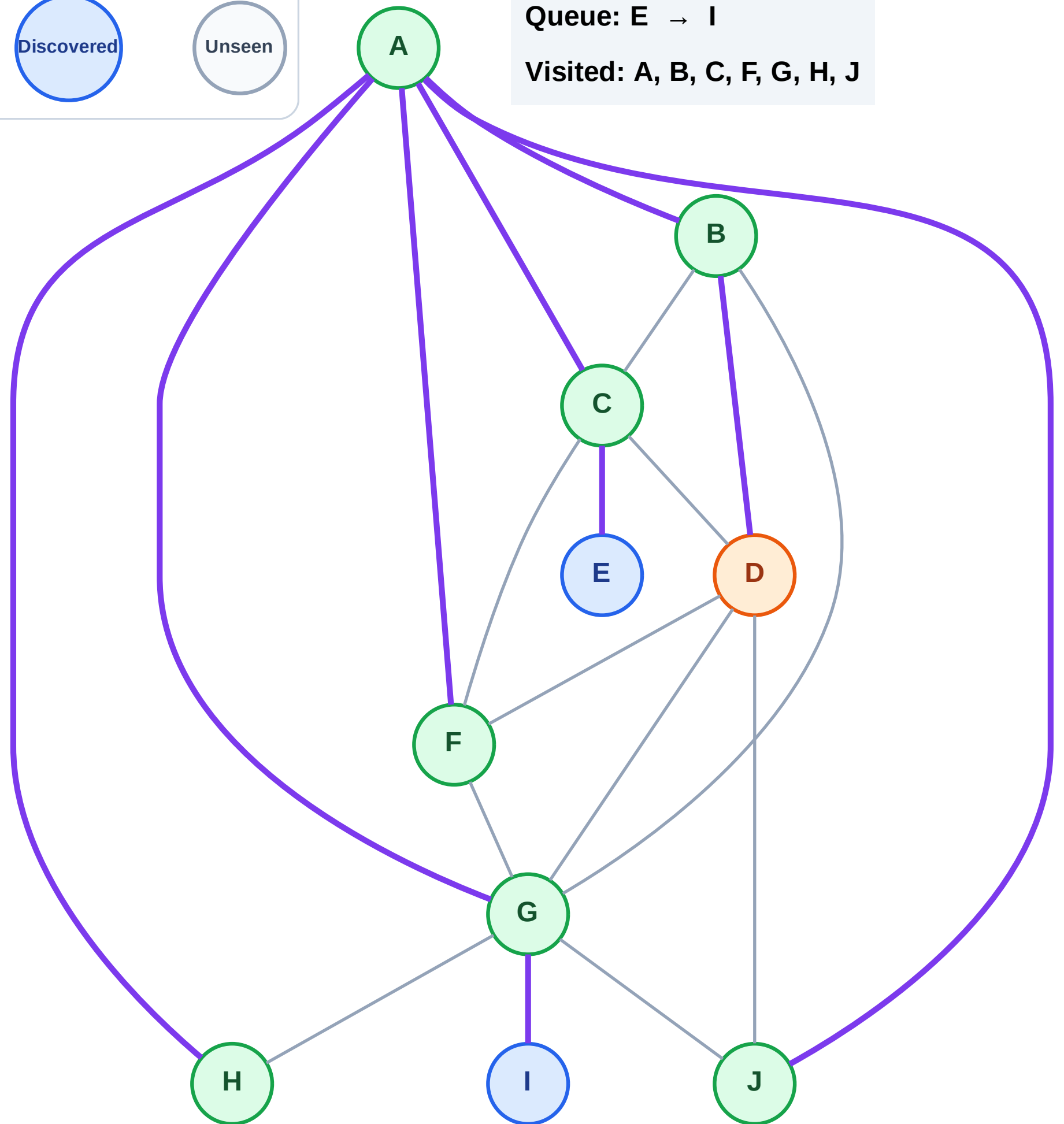
Discovered



Unseen

Queue: E → I

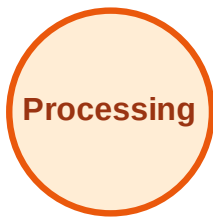
Visited: A, B, C, F, G, H, J



BFS Traversal

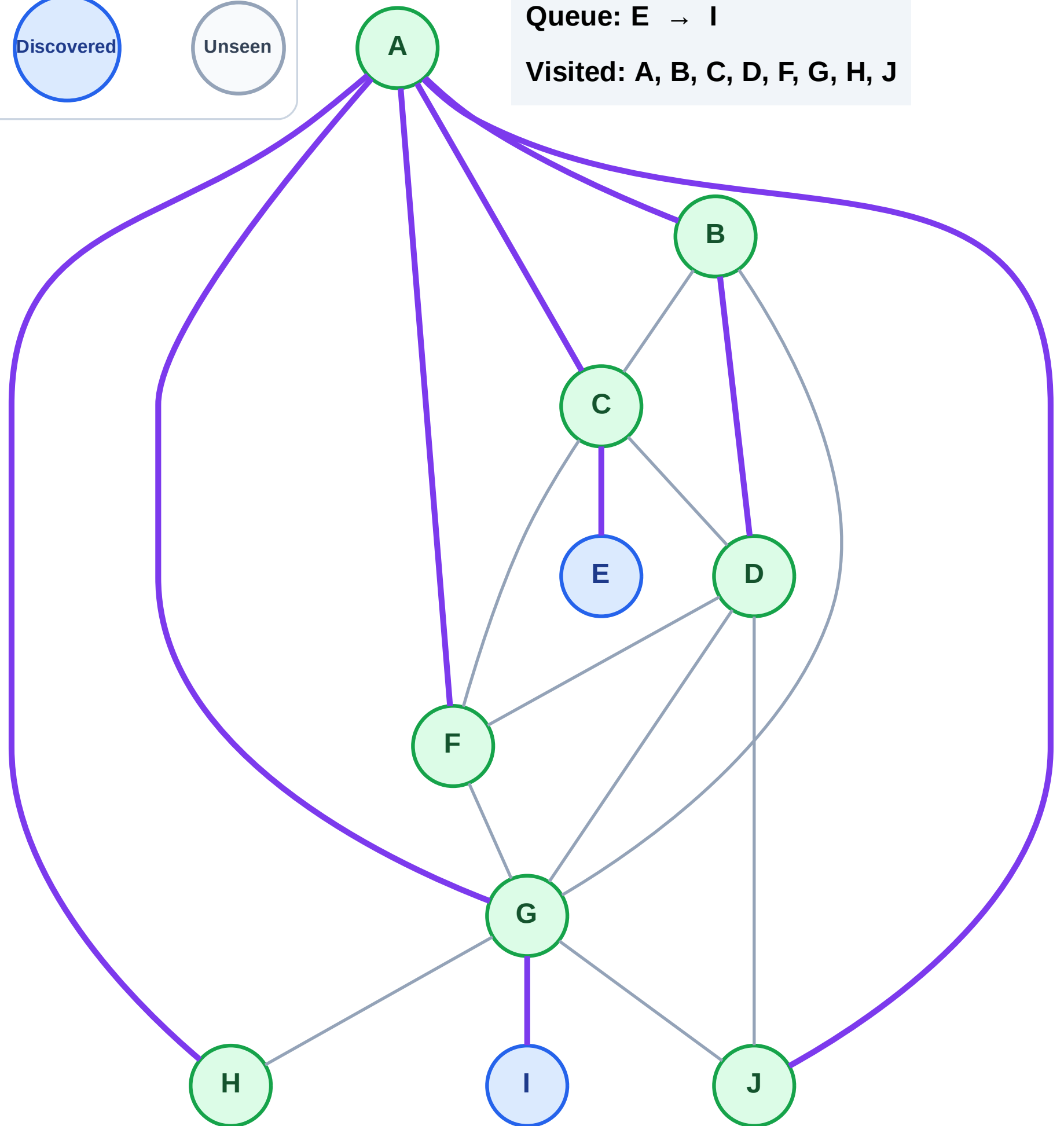
Step 17: No new neighbors from D

Legend



Queue: E → I

Visited: A, B, C, D, F, G, H, J



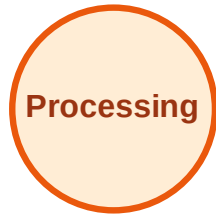
BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



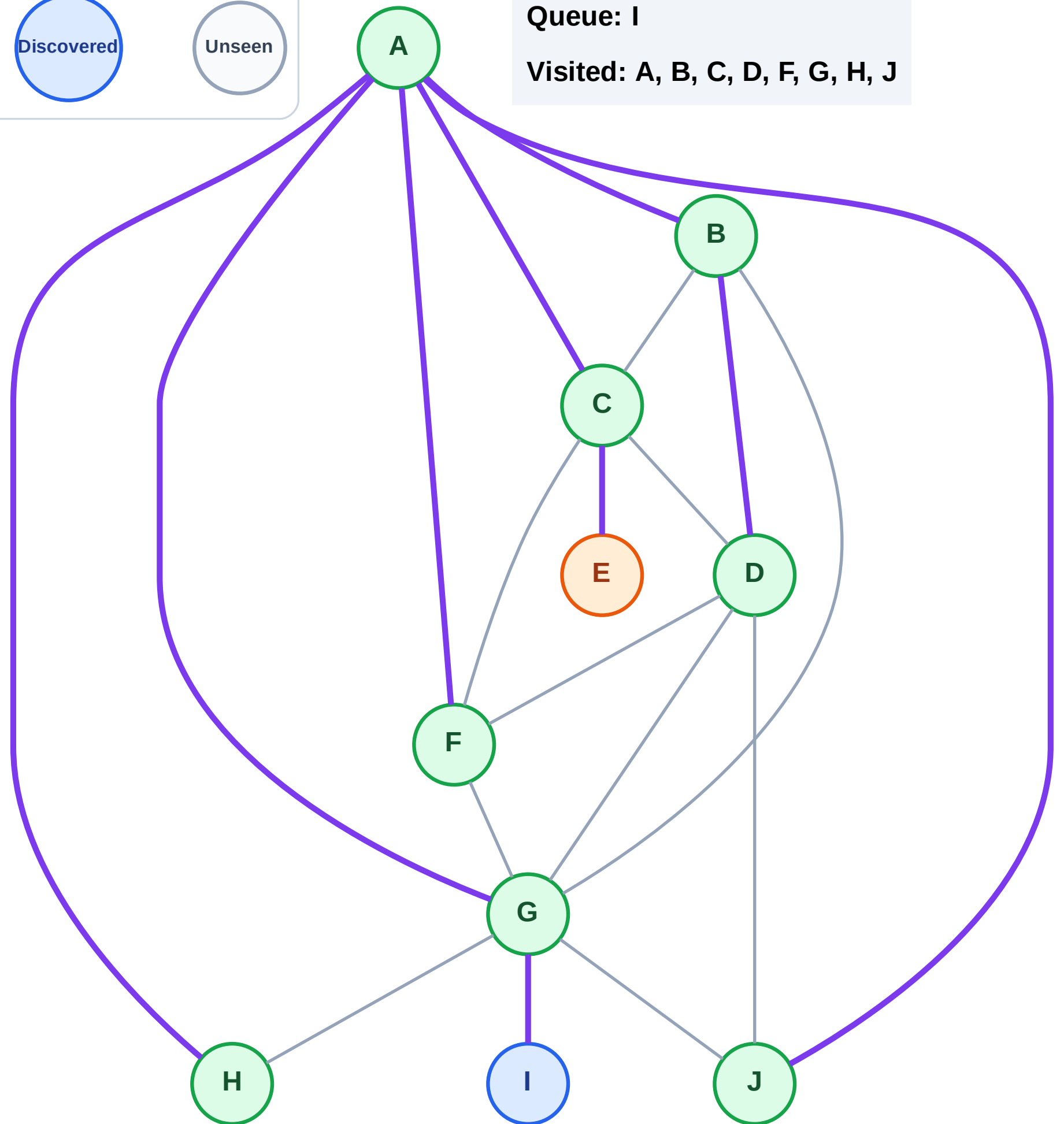
Discovered



Unseen

Queue: I

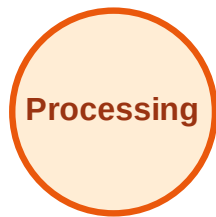
Visited: A, B, C, D, F, G, H, J



BFS Traversal

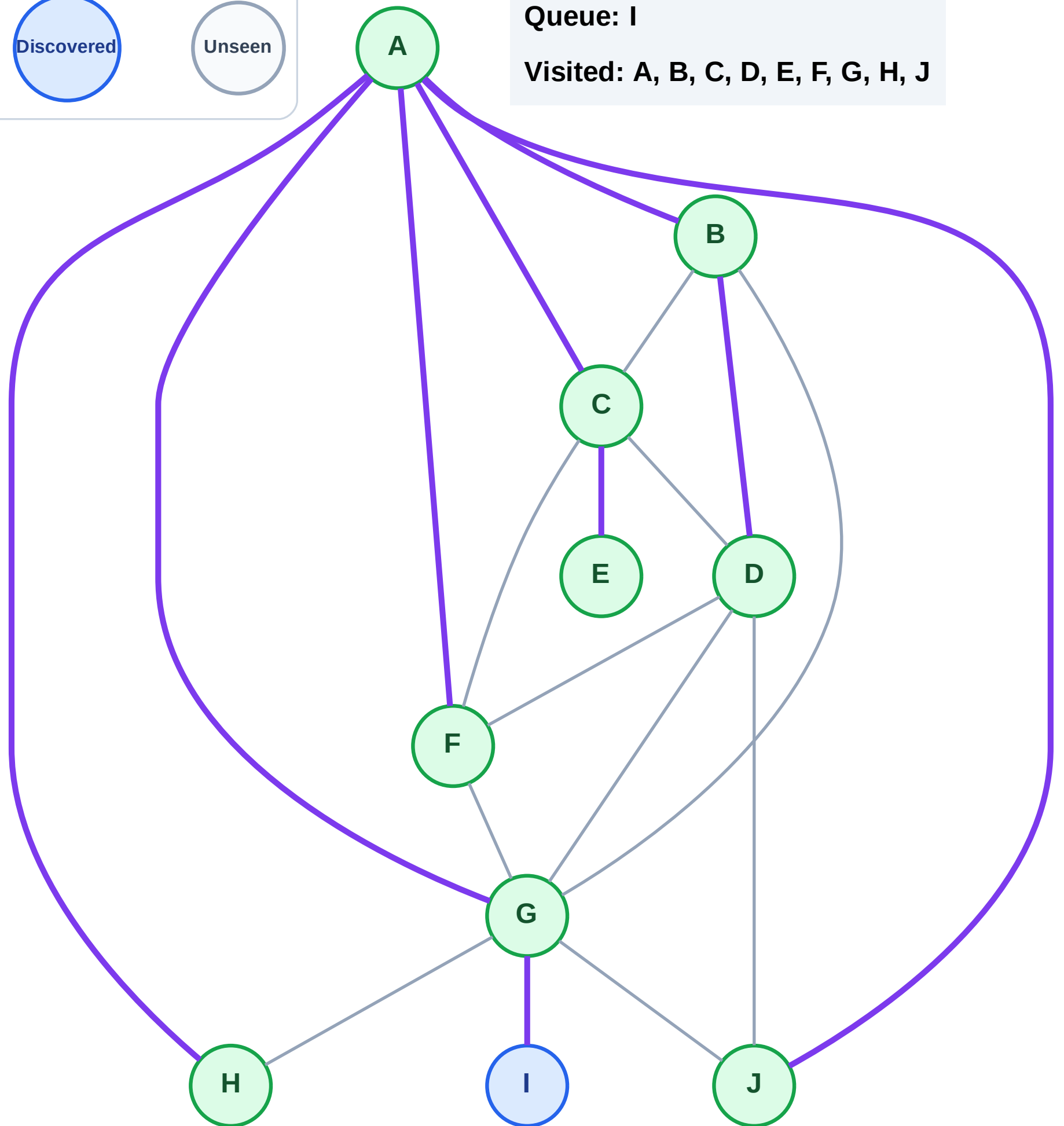
Step 19: No new neighbors from E

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



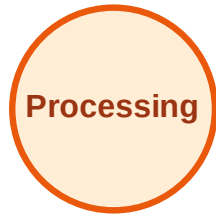
BFS Traversal

Step 20: Process node I

Legend



Complete



Processing



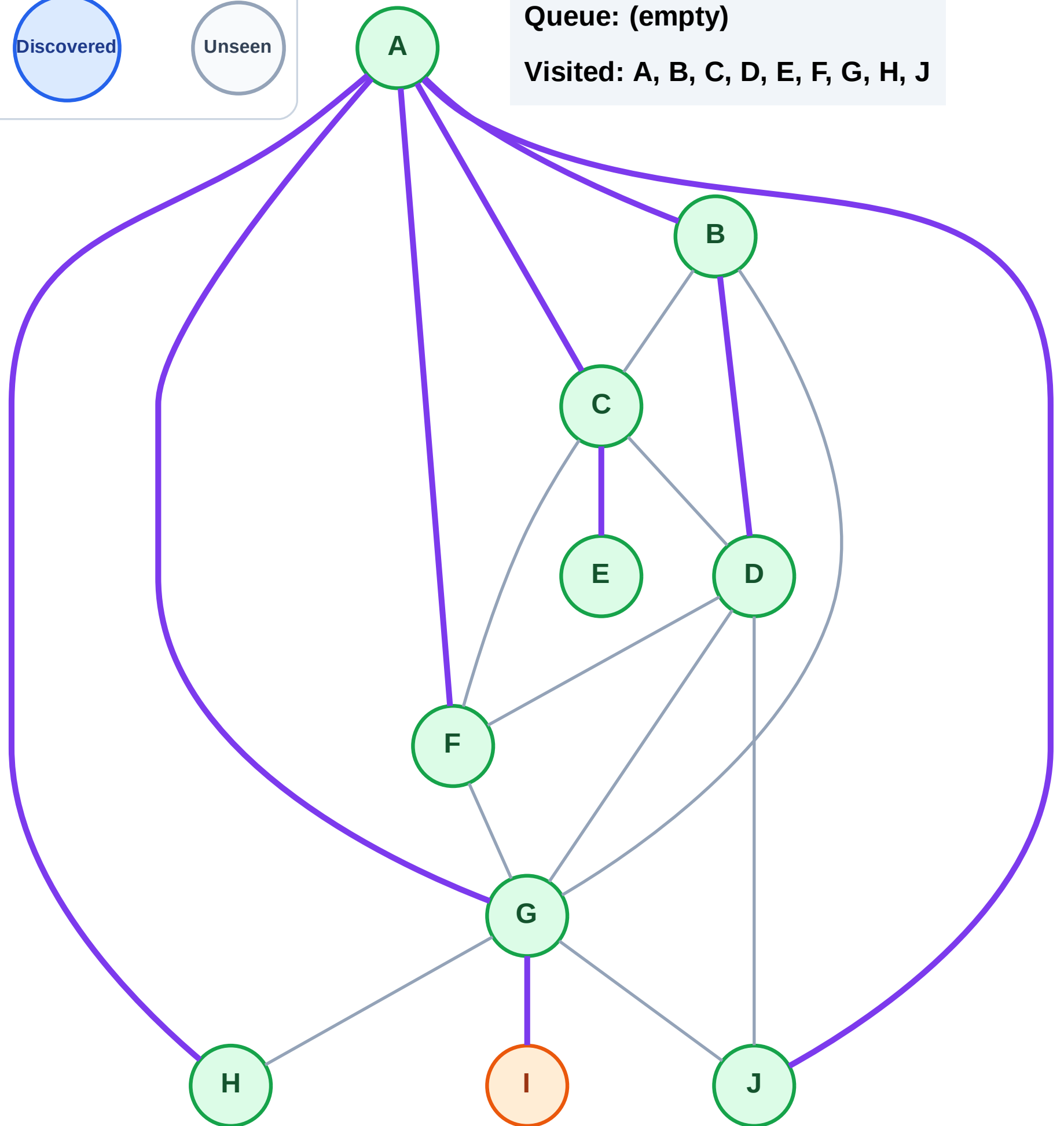
Discovered



Unseen

Queue: (empty)

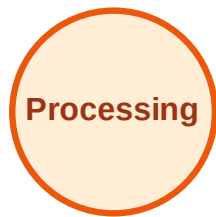
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

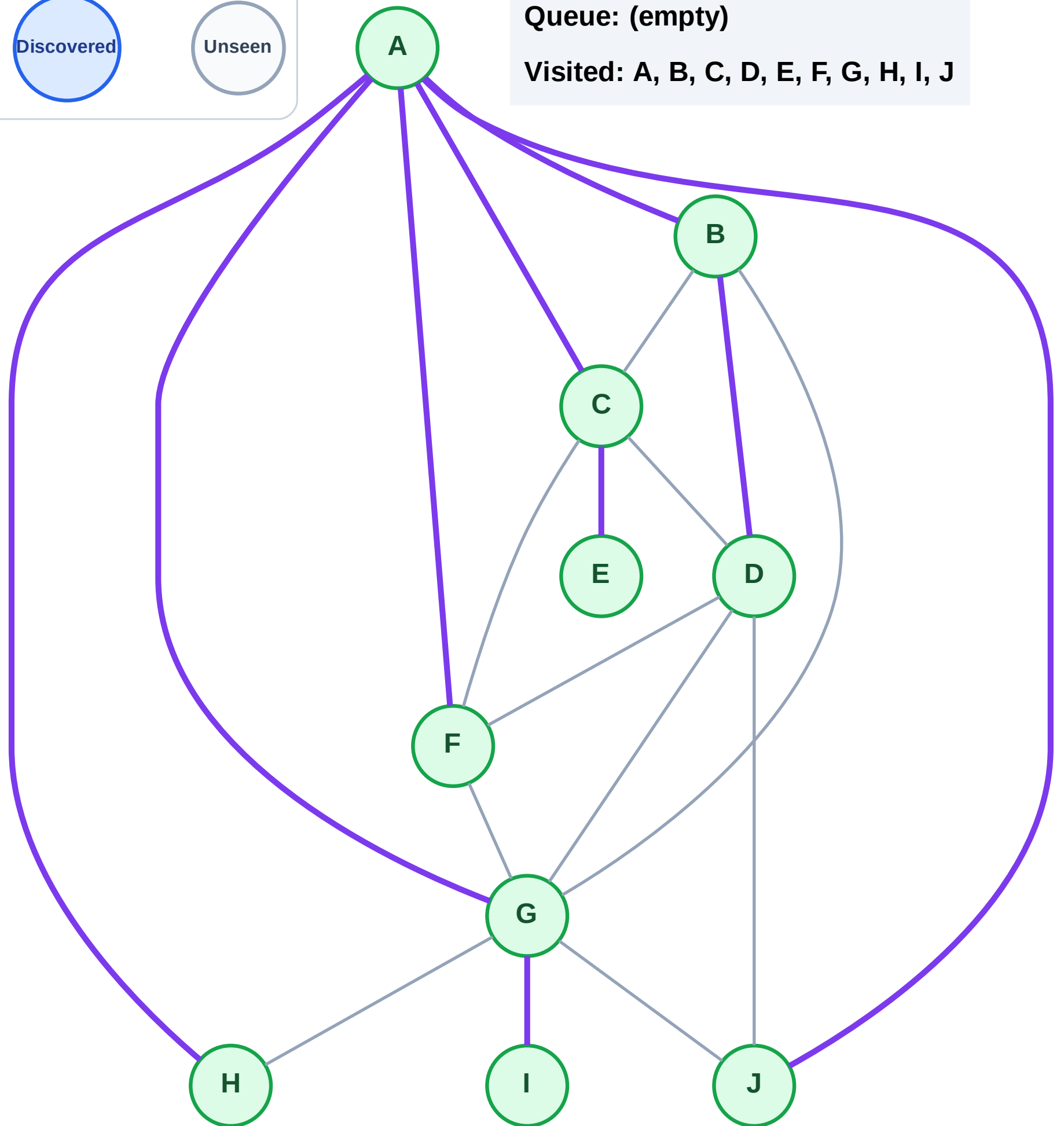
Step 21: No new neighbors from I

Legend



Queue: (empty)

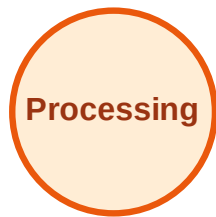
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

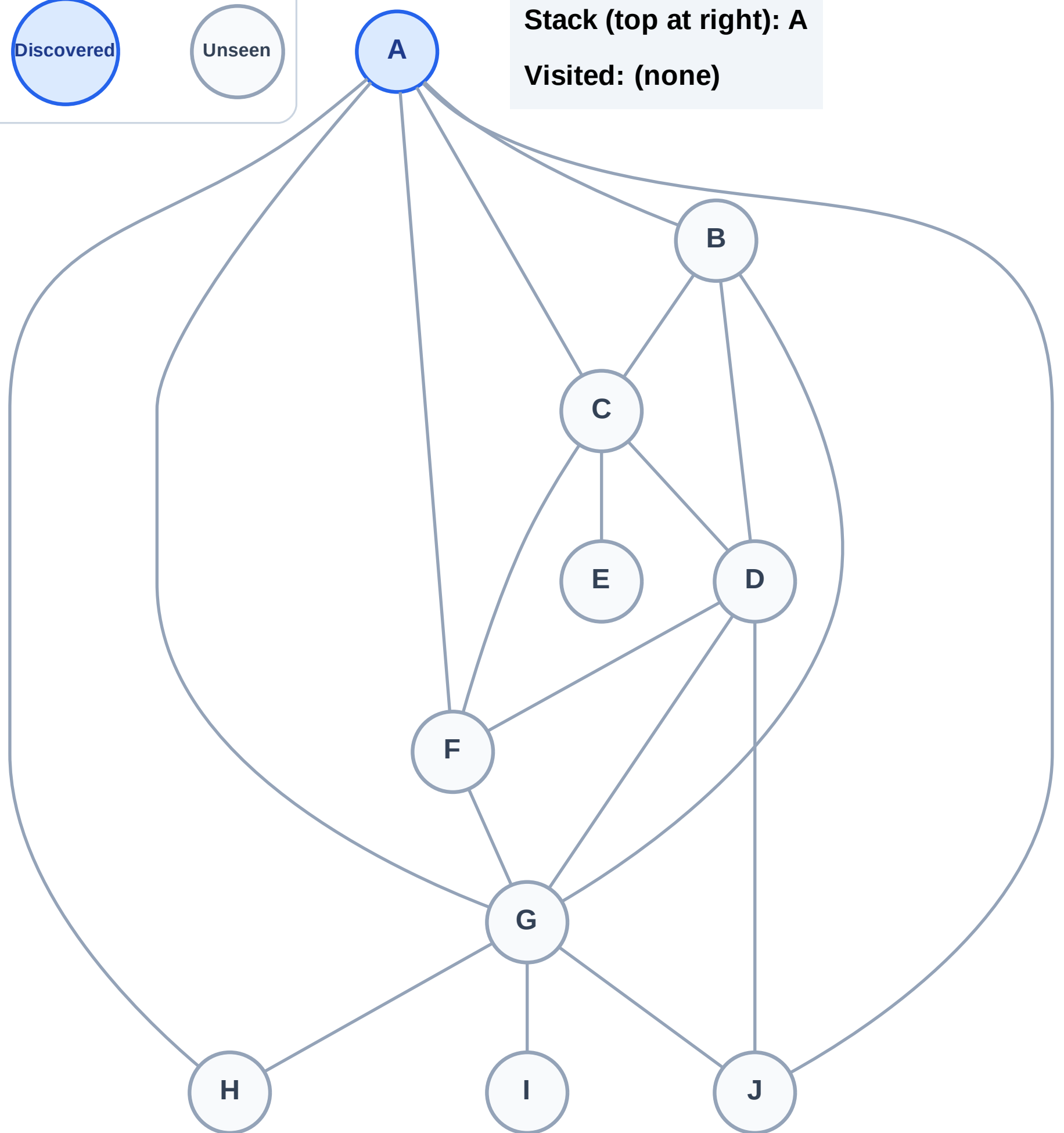
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



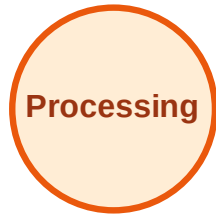
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



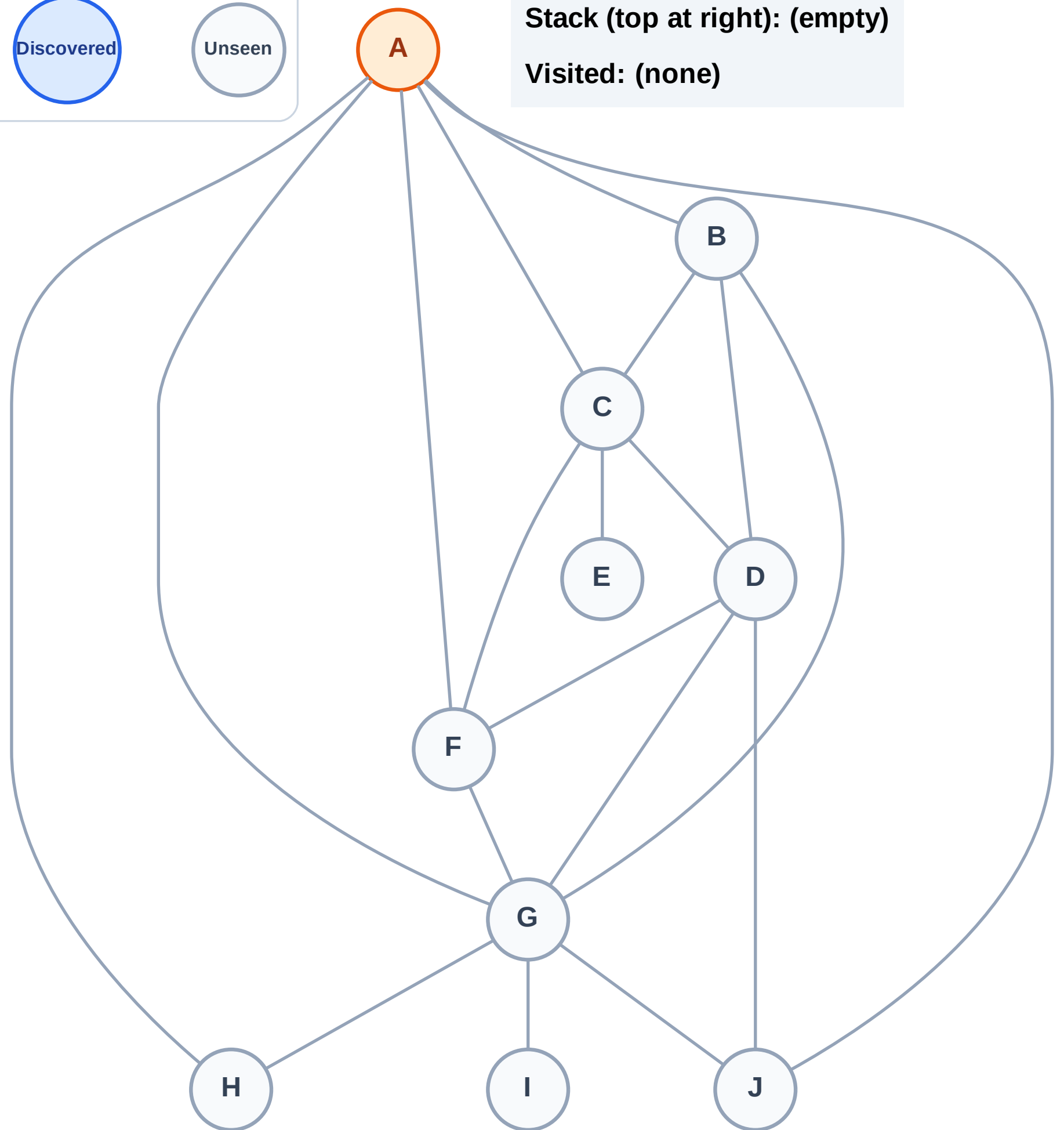
Discovered



Unseen

Stack (top at right): (empty)

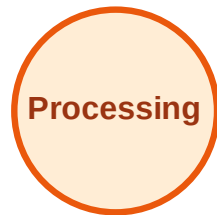
Visited: (none)



DFS Traversal

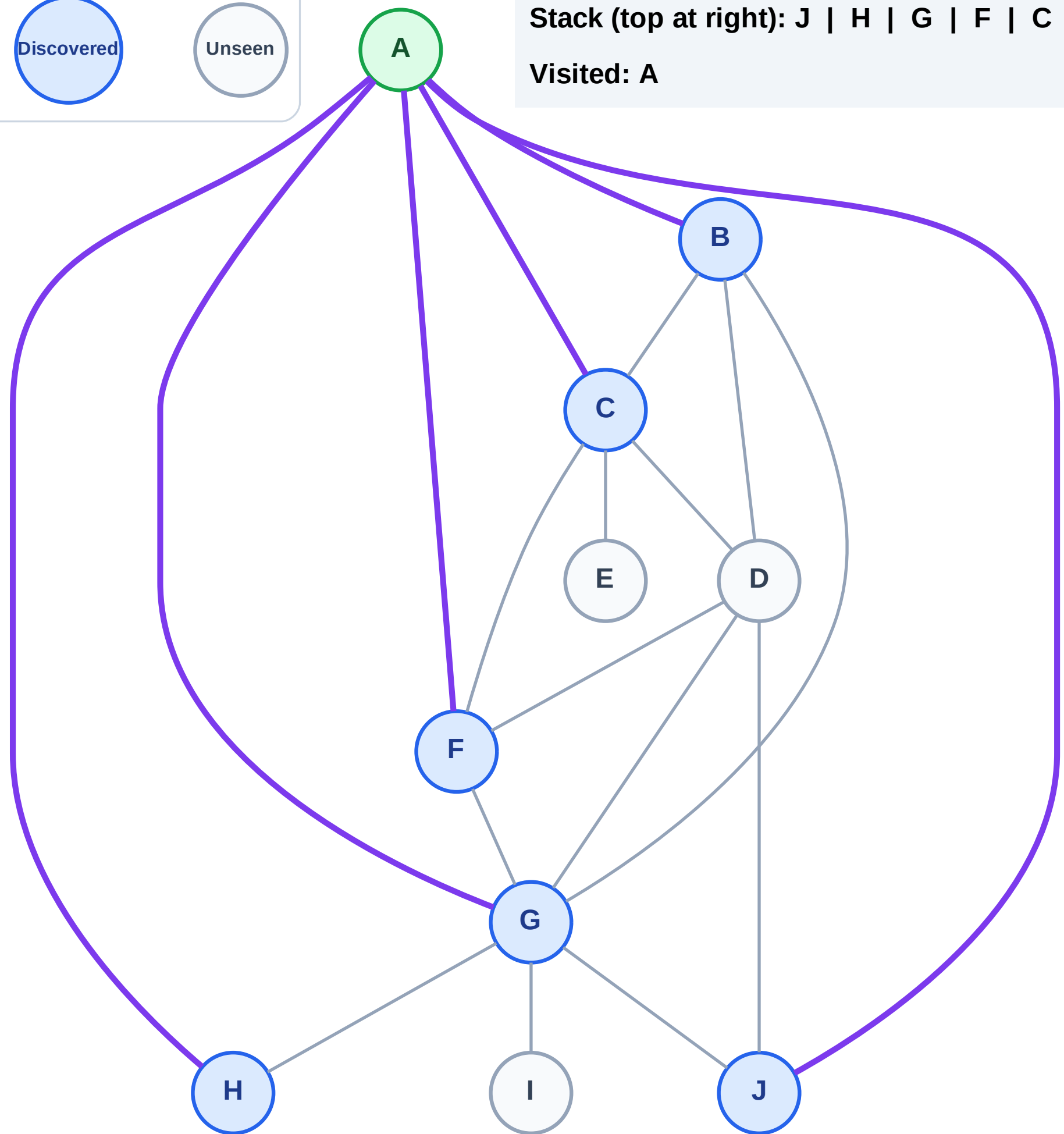
Step 3: Discover J, H, G, F, C, B from A

Legend



Stack (top at right): J | H | G | F | C | B

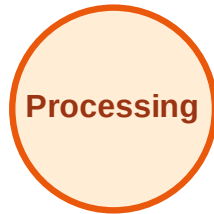
Visited: A



DFS Traversal

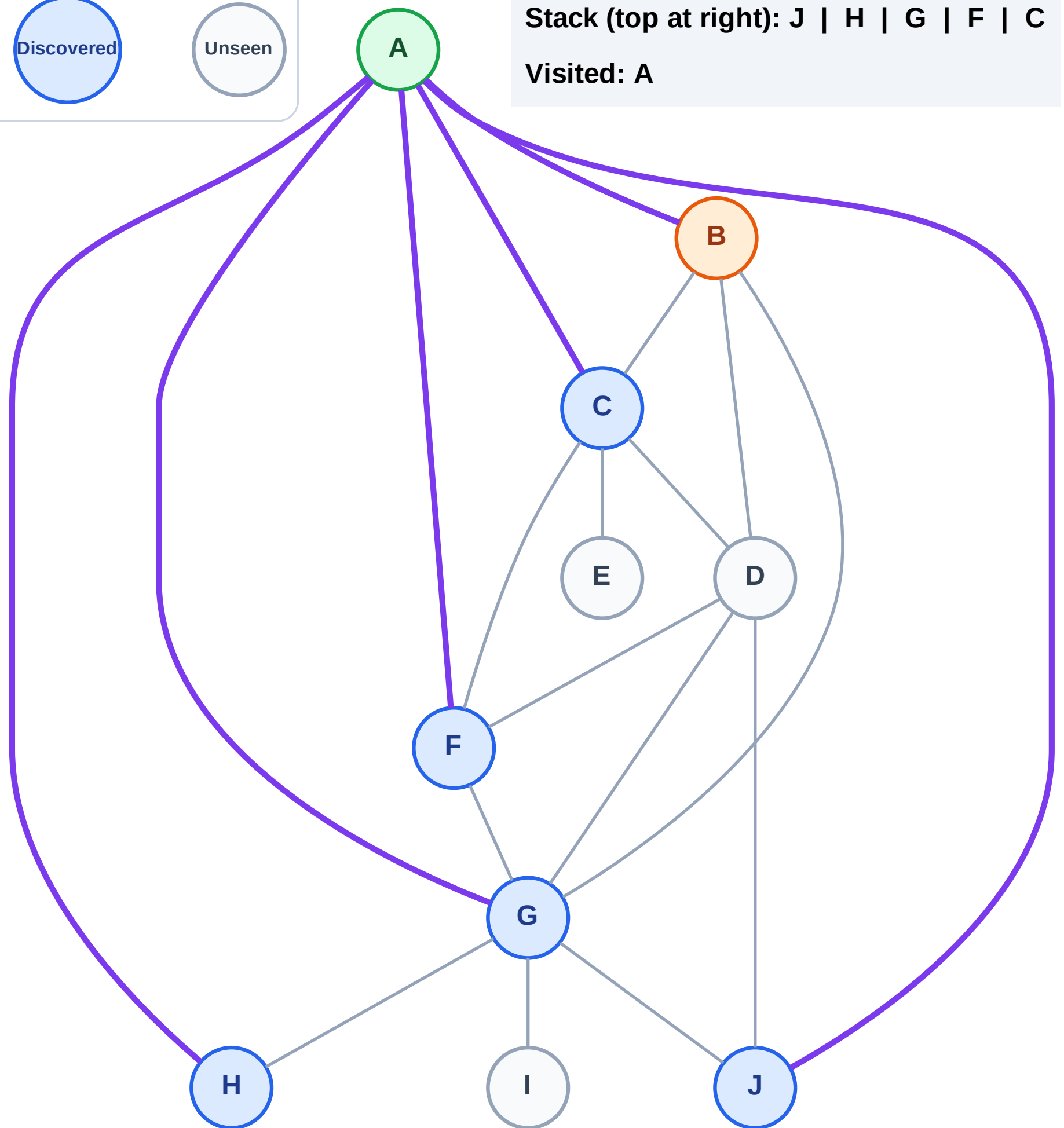
Step 4: Process node B

Legend



Stack (top at right): J | H | G | F | C

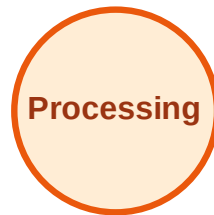
Visited: A



DFS Traversal

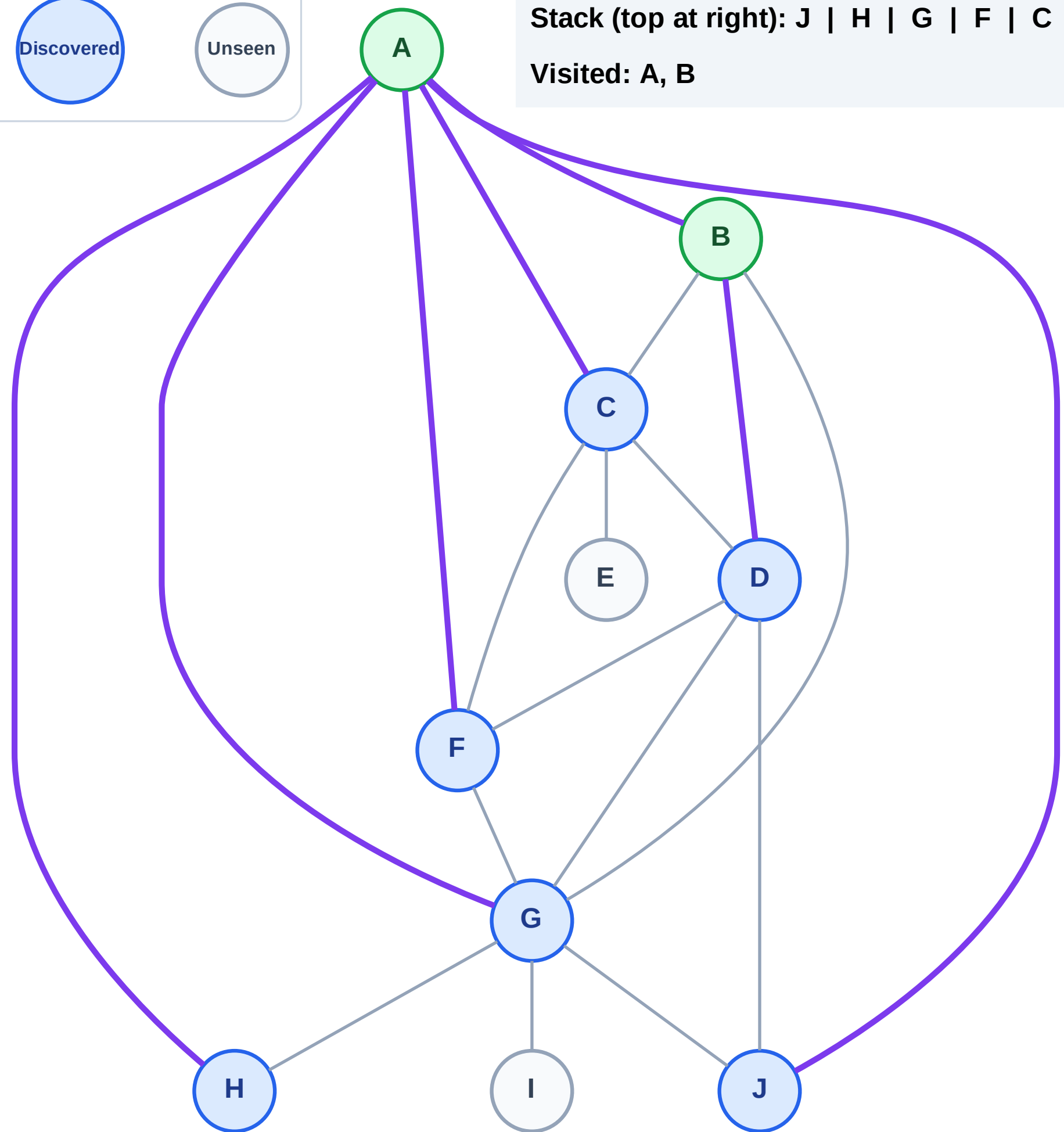
Step 5: Discover D from B

Legend



Stack (top at right): J | H | G | F | C | D

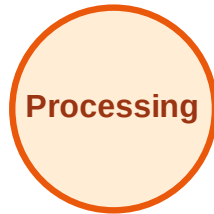
Visited: A, B



DFS Traversal

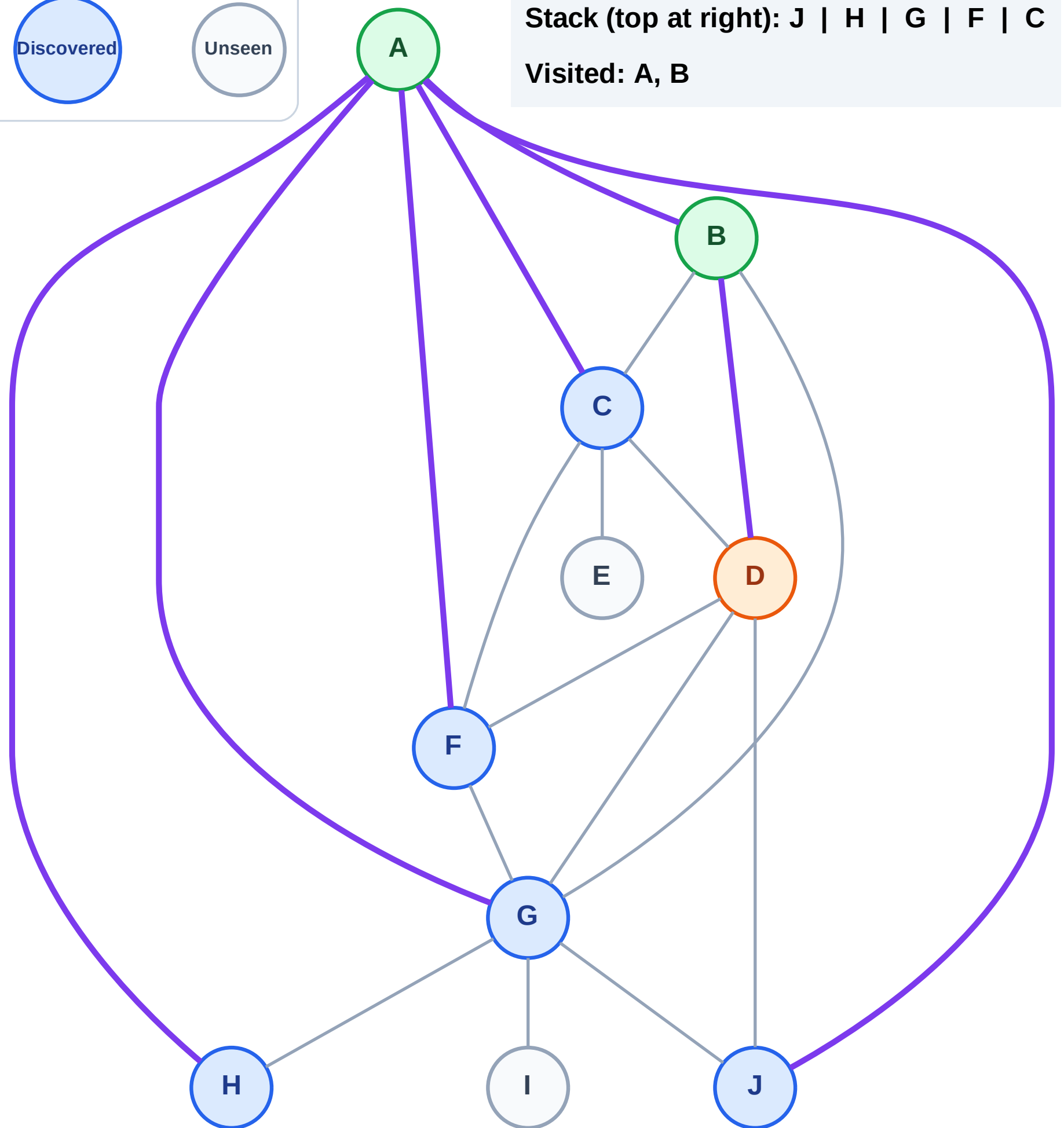
Step 6: Process node D

Legend



Stack (top at right): J | H | G | F | C

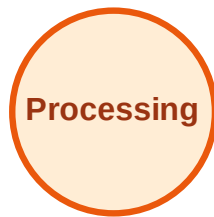
Visited: A, B



DFS Traversal

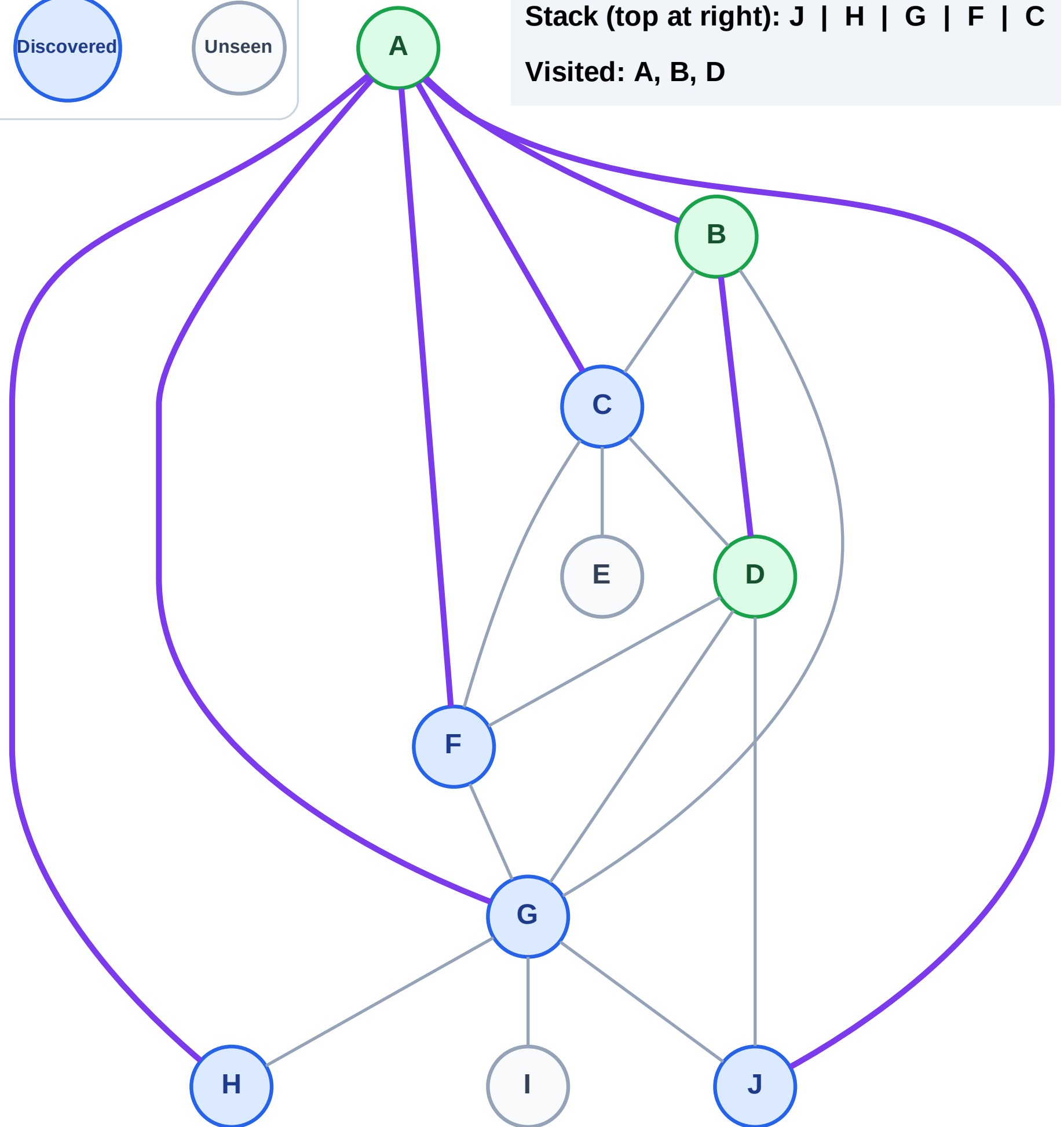
Step 7: No new neighbors from D

Legend



Stack (top at right): J | H | G | F | C

Visited: A, B, D



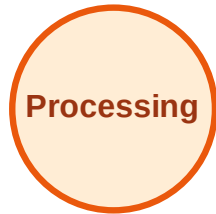
DFS Traversal

Step 8: Process node C

Legend



Complete



Processing



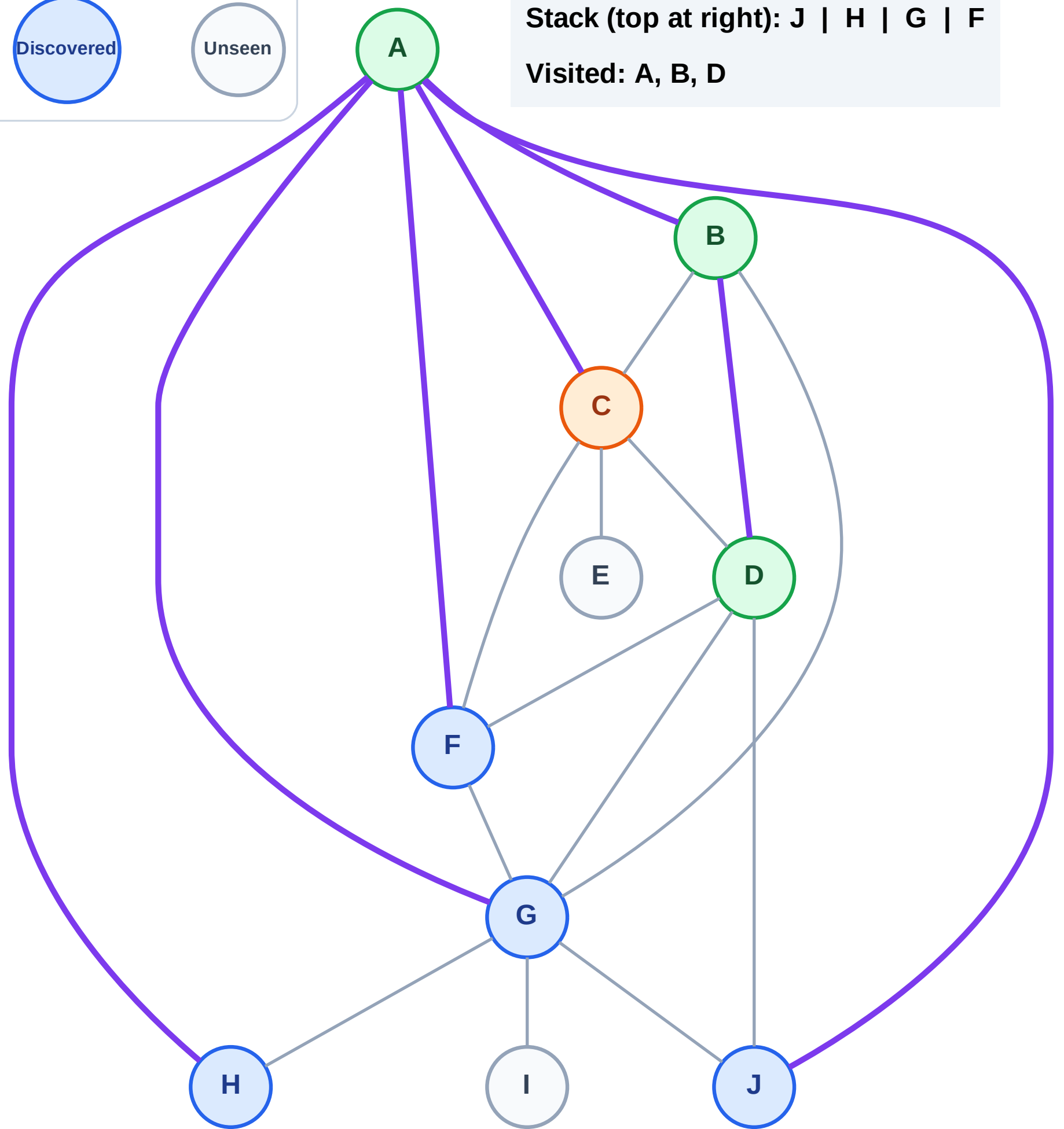
Discovered



Unseen

Stack (top at right): J | H | G | F

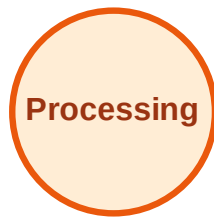
Visited: A, B, D



DFS Traversal

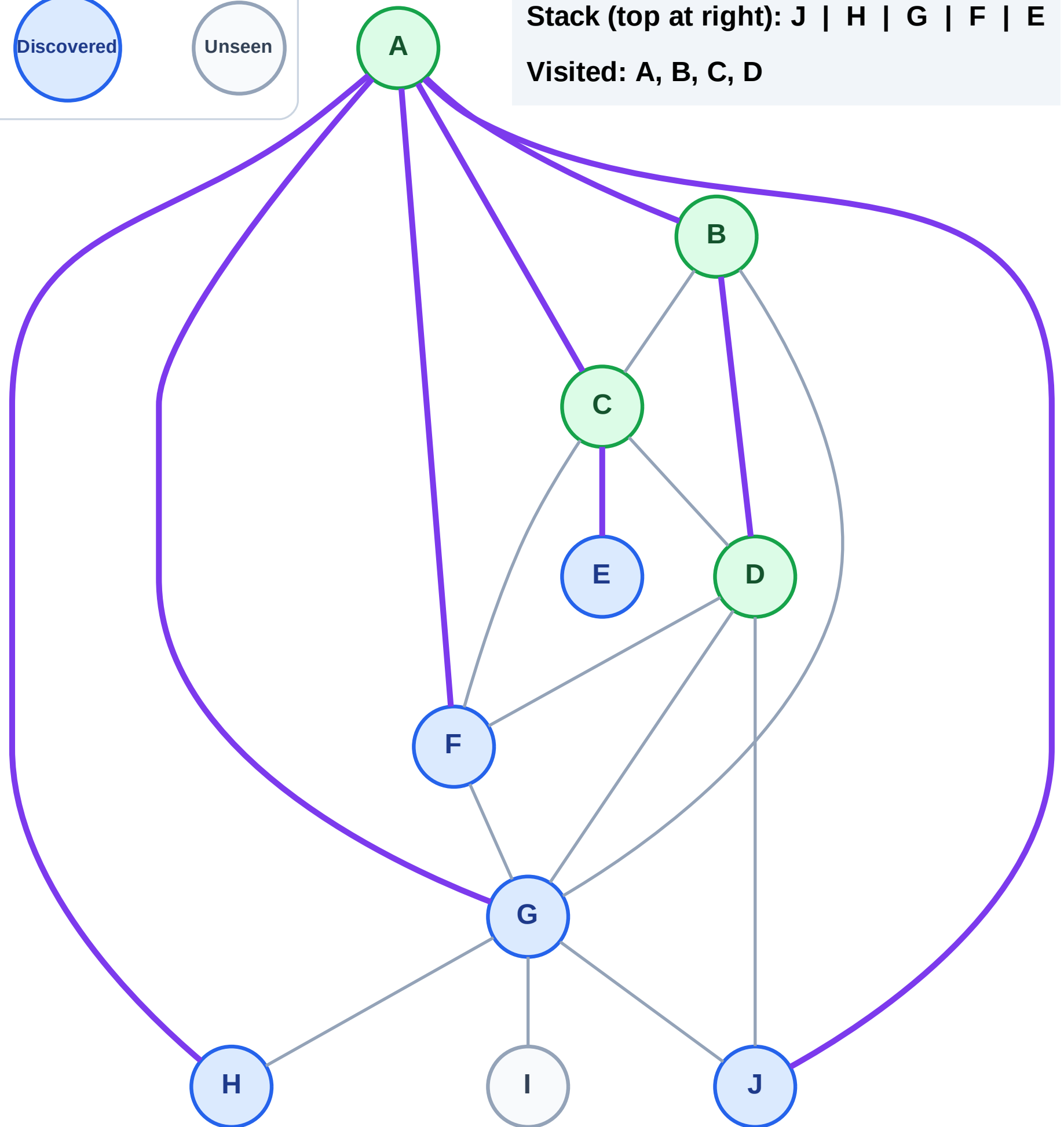
Step 9: Discover E from C

Legend

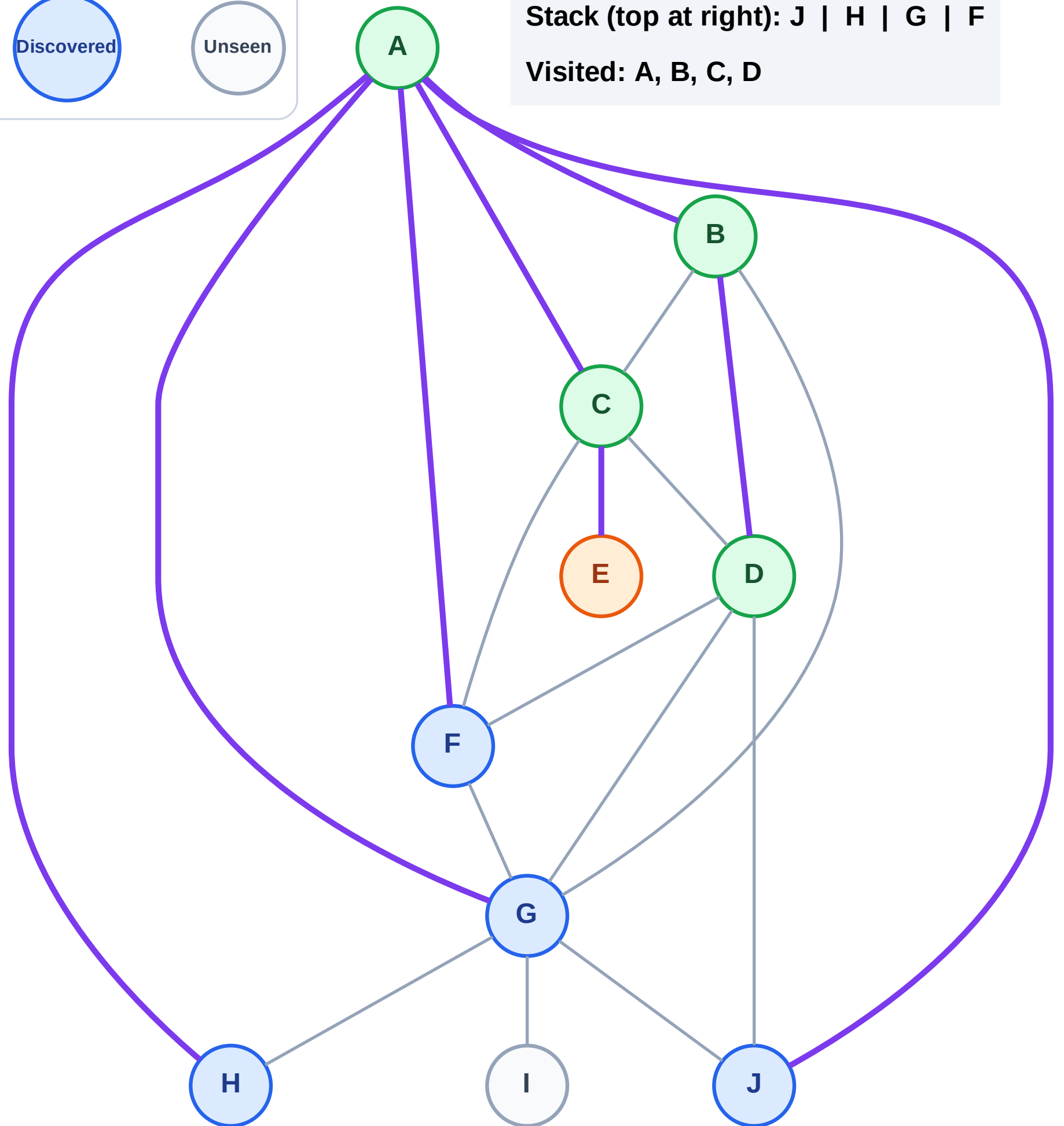
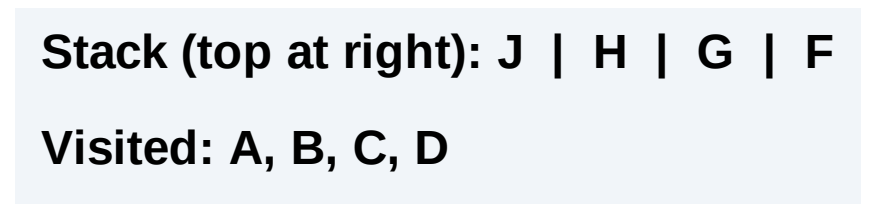


Stack (top at right): J | H | G | F | E

Visited: A, B, C, D



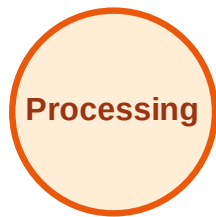
Step 10: Process node E



DFS Traversal

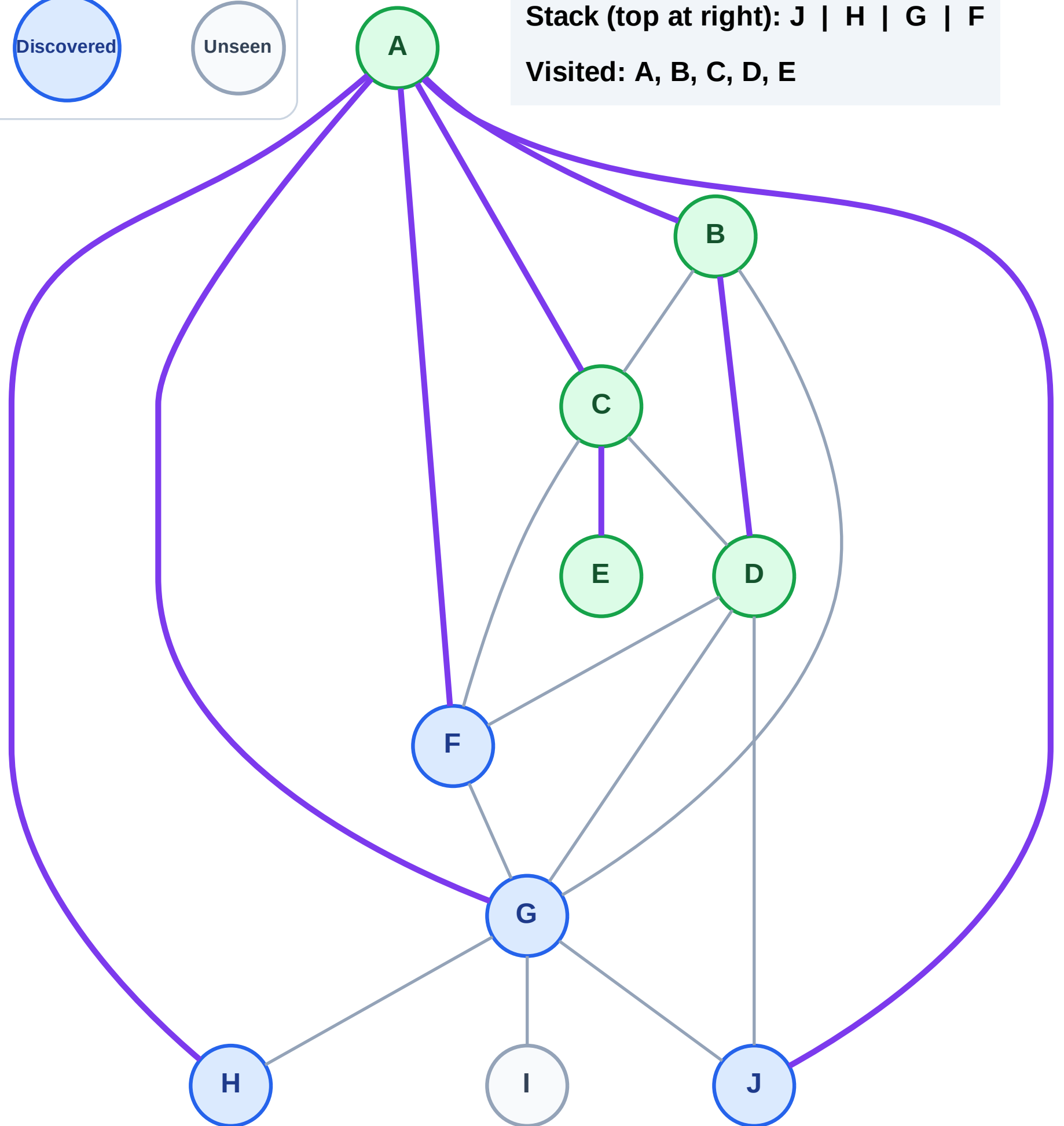
Step 11: No new neighbors from E

Legend



Stack (top at right): J | H | G | F

Visited: A, B, C, D, E



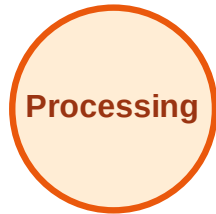
DFS Traversal

Step 12: Process node F

Legend



Complete



Processing



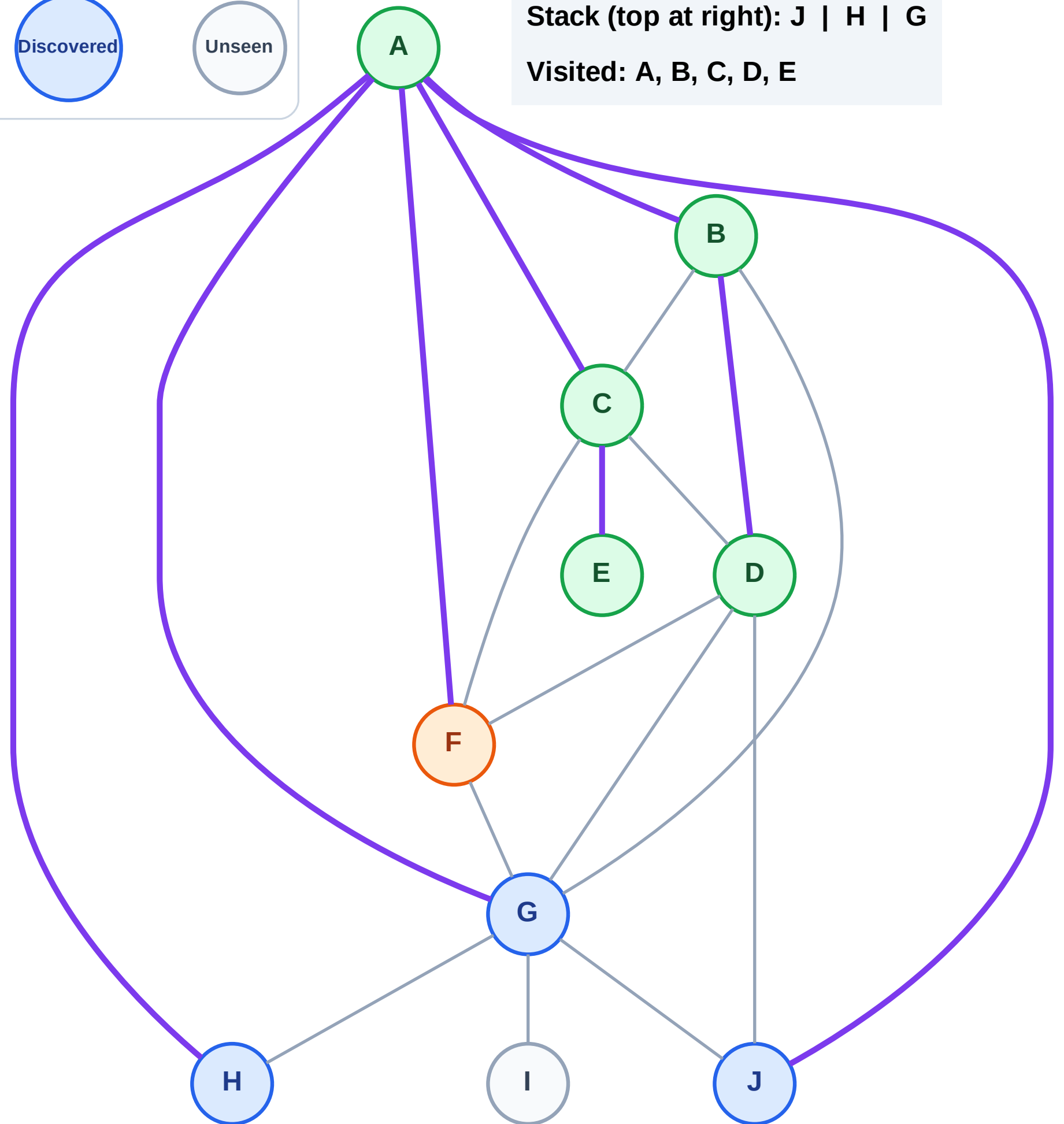
Discovered



Unseen

Stack (top at right): J | H | G

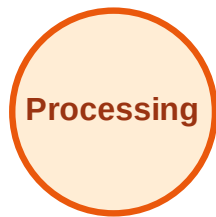
Visited: A, B, C, D, E



DFS Traversal

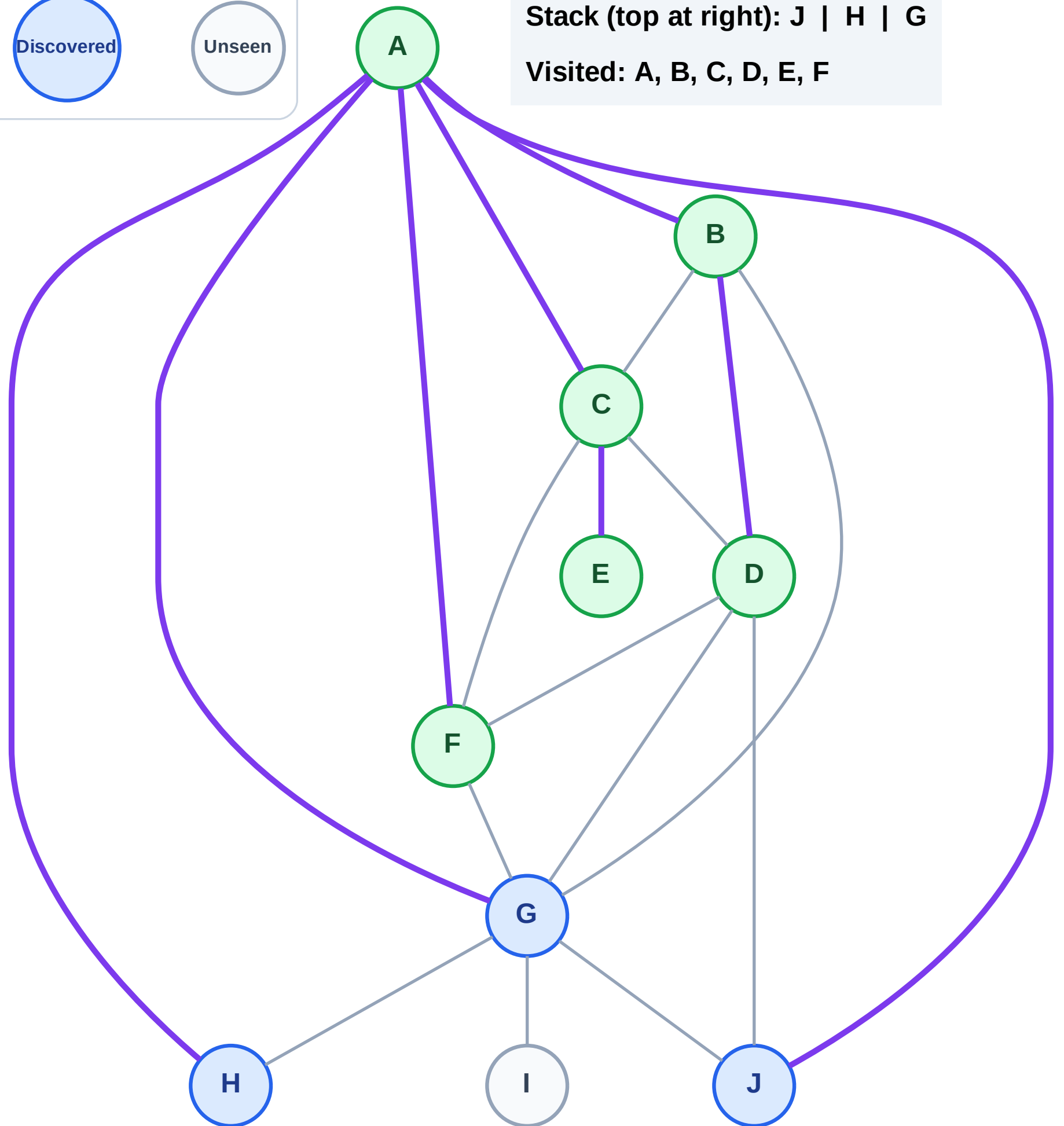
Step 13: No new neighbors from F

Legend



Stack (top at right): J | H | G

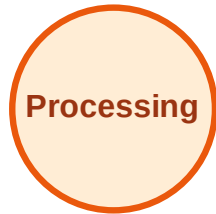
Visited: A, B, C, D, E, F



DFS Traversal

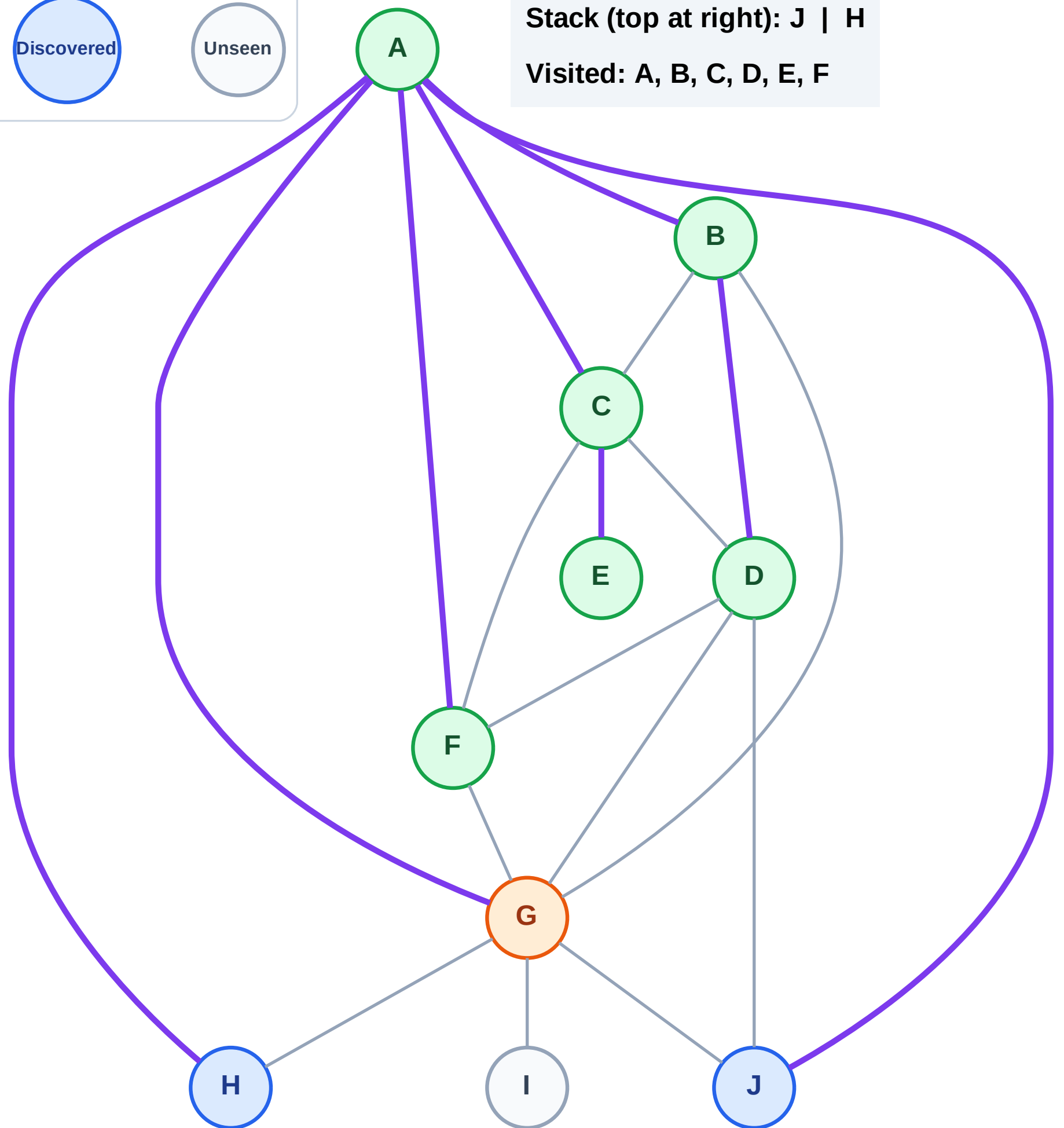
Step 14: Process node G

Legend



Stack (top at right): J | H

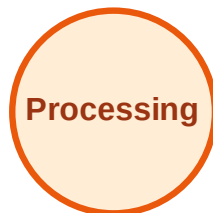
Visited: A, B, C, D, E, F



DFS Traversal

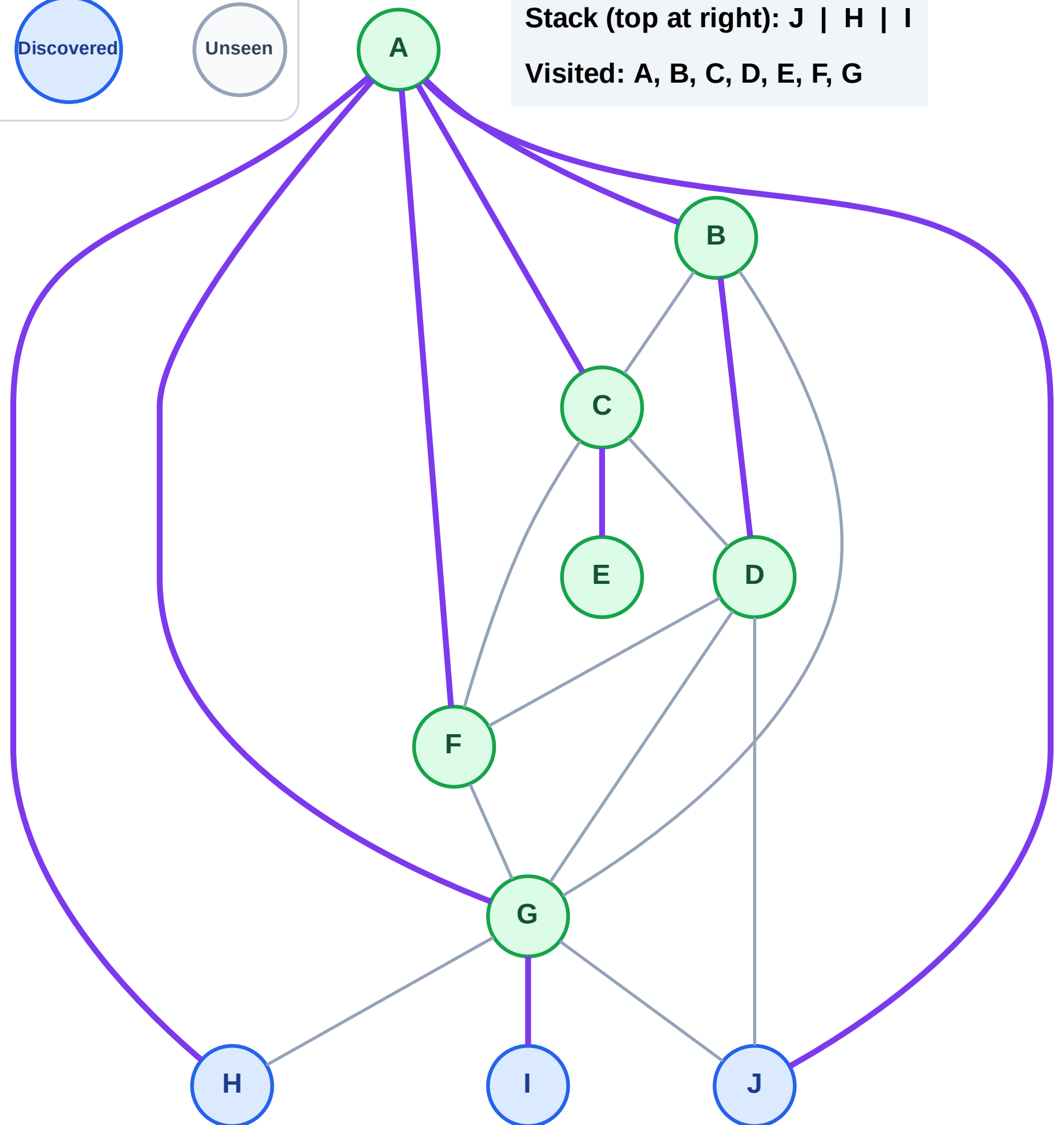
Step 15: Discover I from G

Legend



Stack (top at right): J | H | I

Visited: A, B, C, D, E, F, G



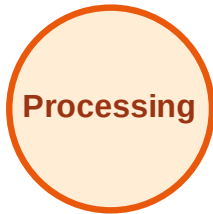
DFS Traversal

Step 16: Process node I

Legend



Complete



Processing



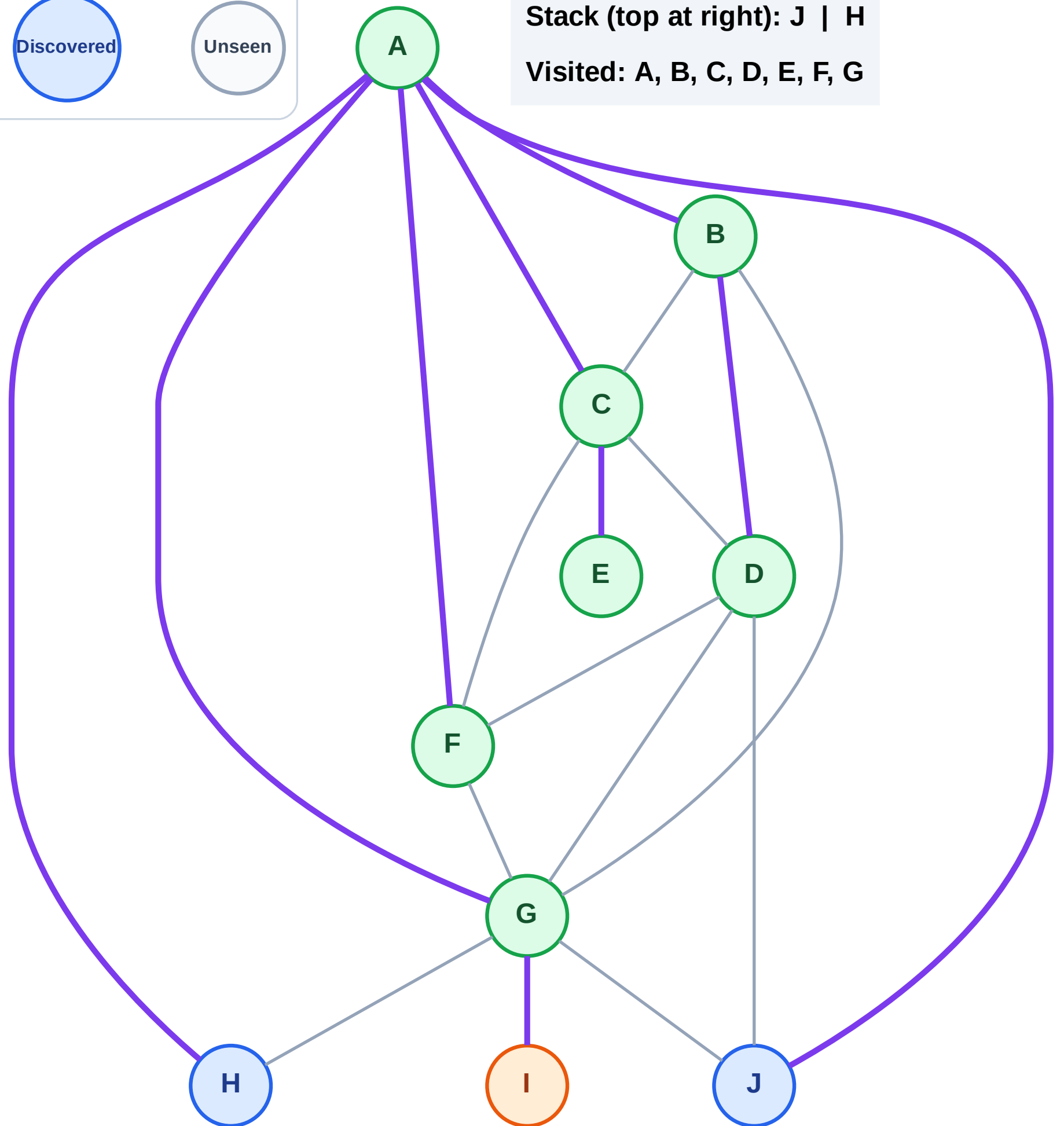
Discovered



Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G



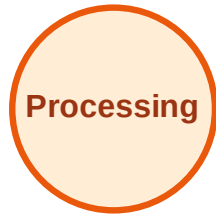
DFS Traversal

Step 17: No new neighbors from I

Legend



Complete



Processing



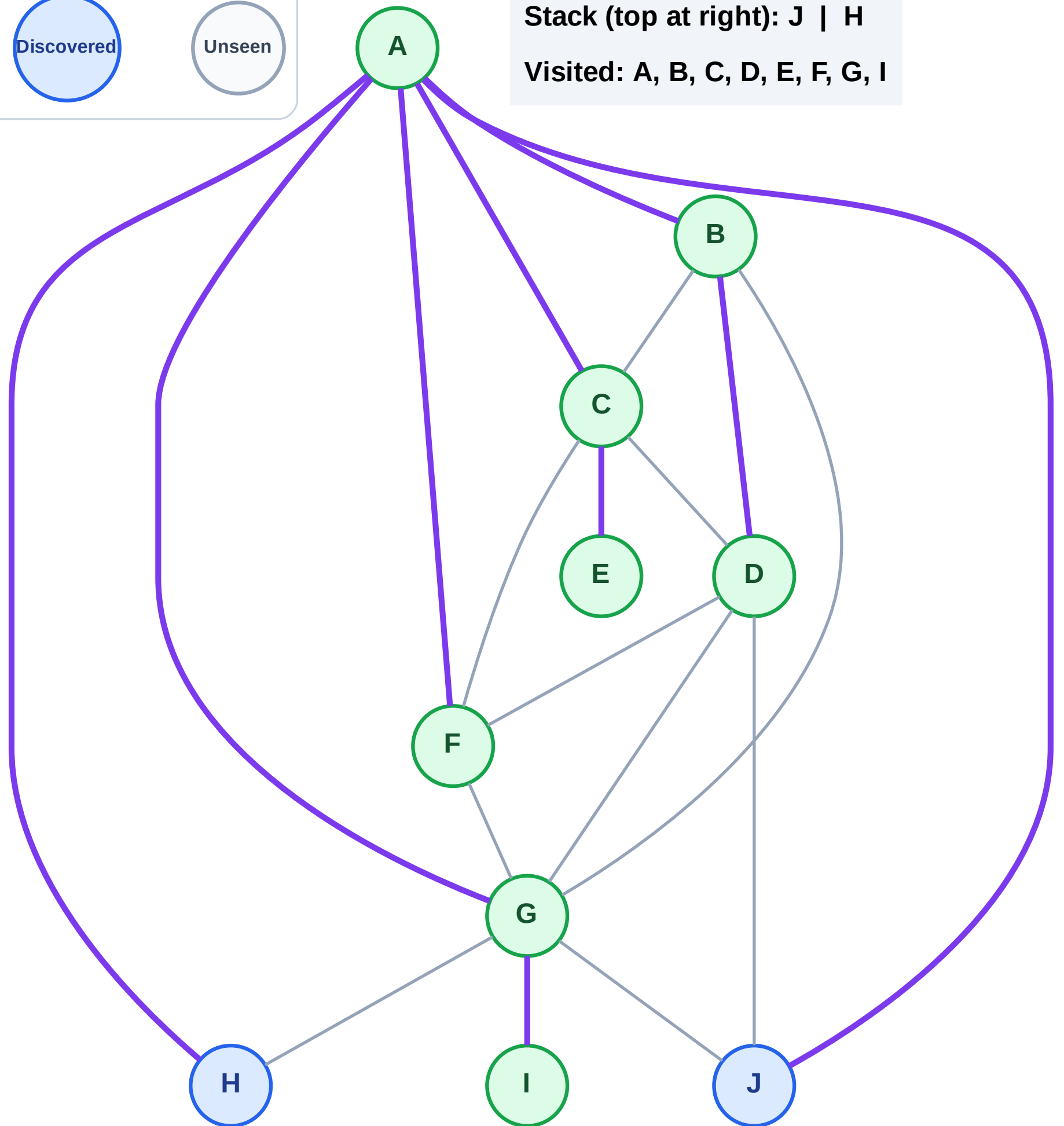
Discovered



Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

Complete

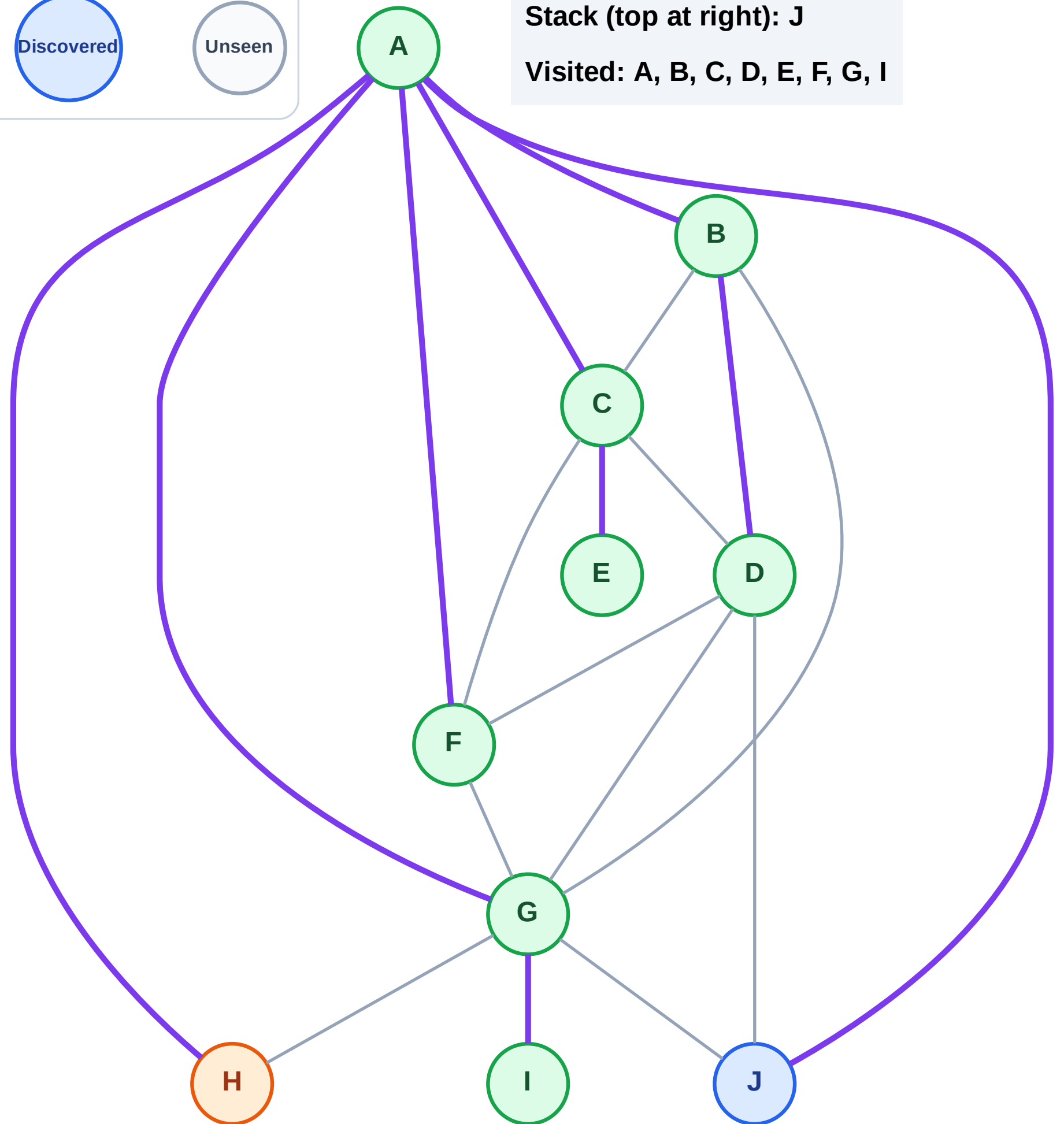
Processing

Discovered

Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, I



DFS Traversal

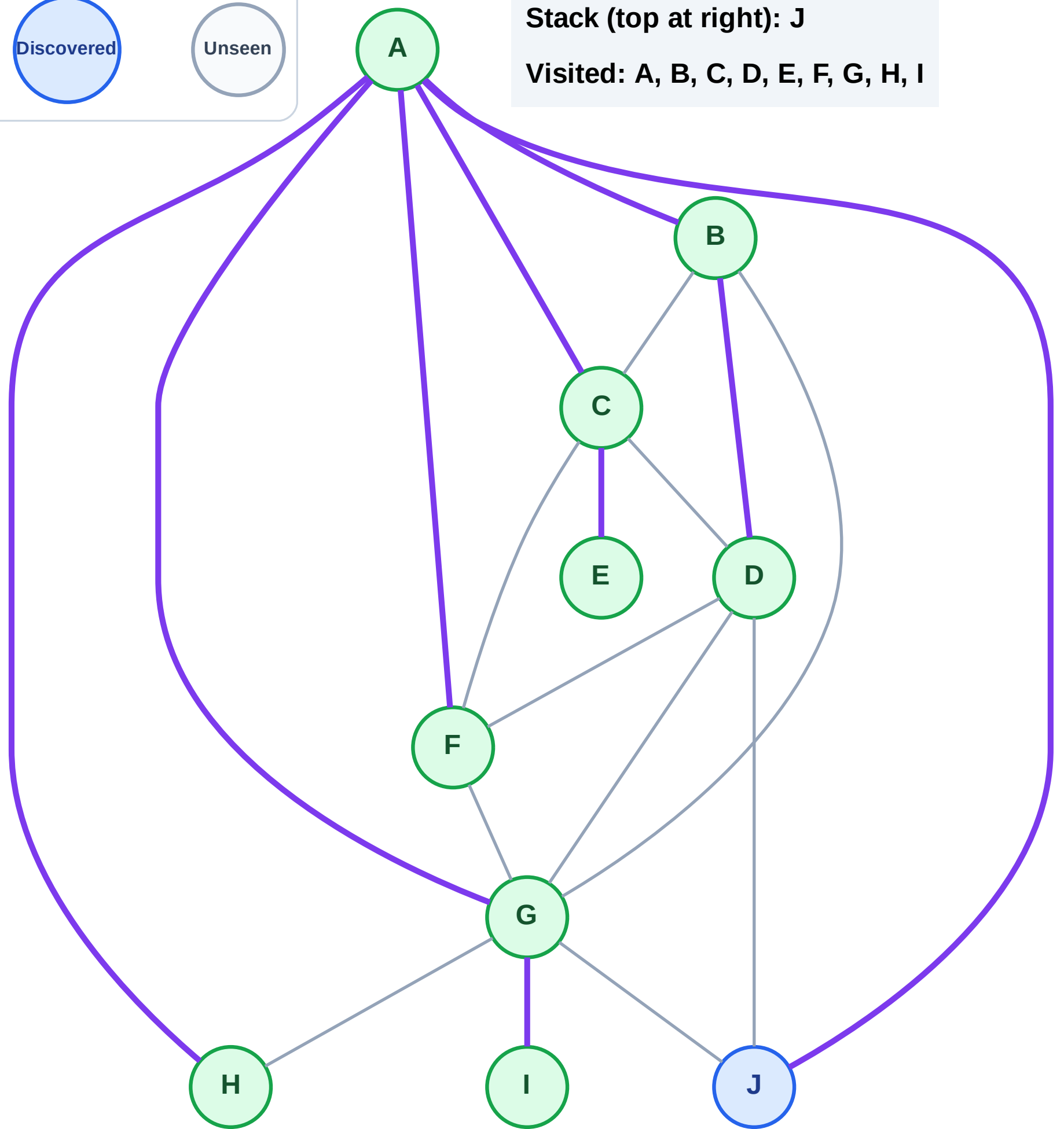
Step 19: No new neighbors from H

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



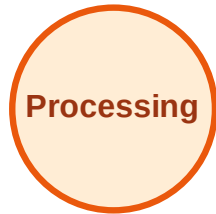
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



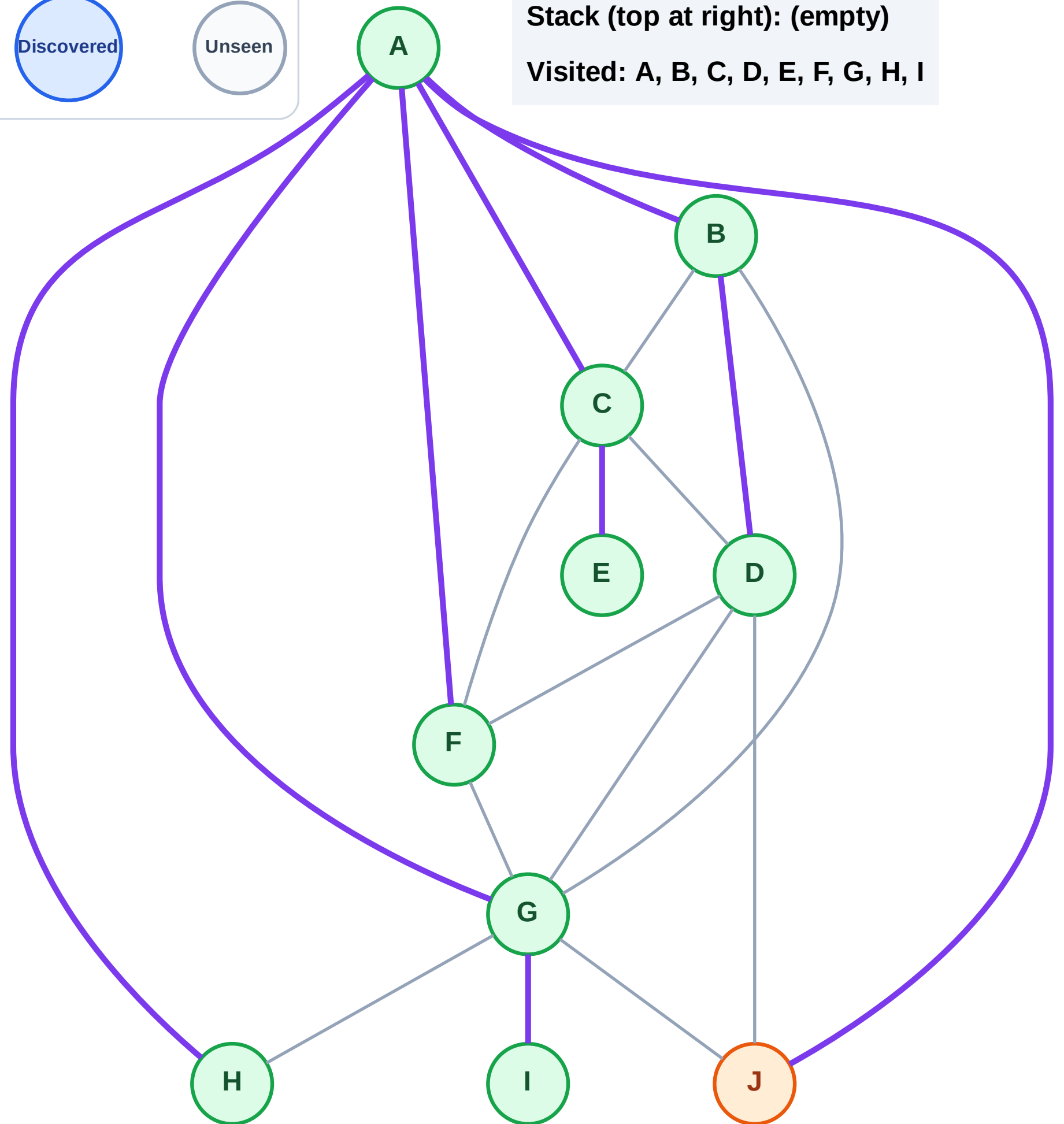
Discovered



Unseen

Stack (top at right): (empty)

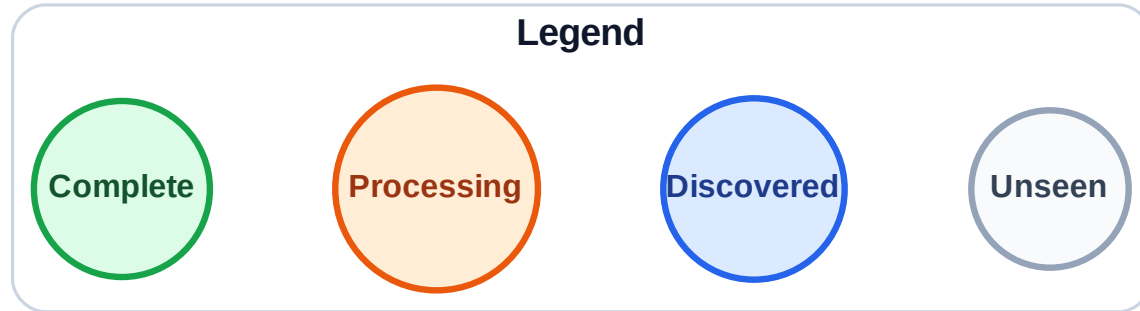
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Step 21: No new neighbors from J



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

